Artificial Intelligence Review (2026) 59:72
https://doi.org/10.1007/s10462-025-11470-w

A survey on large language models driven meta-optimizers for automated intelligent optimization

Yan Zheng[1] · Lida Zhang[1] · Kaiwen Li[1] · Rui Wang[1] · Wenhua Li[1] · Tao Zhang[1] · Qingfu Zhang[2] · Yaochu Jin[3]

Received: 23 August 2025 / Accepted: 14 December 2025 / Published online: 6 January 2026
© The Author(s) 2026

Abstract
This review systematically summarizes the research progress of Large Language Models (LLMs) as meta-optimizers in the field of automated intelligent optimization algorithm design, aiming to establish a unified framework for this emerging research direction. The study first defines the core paradigm and research scope of LLM-driven meta-optimization, explaining how it overcomes the limitations of traditional optimization algorithms in terms of parameter sensitivity and cross-domain generalization. It then systematically summarizes representative methodological frameworks and typical methods for key stages such as algorithm auto-generation, dynamic selection, parameter configuration, and mutation control using LLMs, revealing the potential for synergies between generative AI and gradient-free optimization heuristics. Additionally, this paper integrates relevant performance evaluation metrics and benchmarking problems, providing practical references for researchers. Finally, the paper discusses the main challenges in this field and outlines future research directions, emphasizing the core role of LLMs as meta-optimizers in driving paradigm shifts toward algorithm design automation and intelligence.

Keywords Large language models · Meta optimization · Automated algorithm design · Evolutionary computation · Intelligent optimization

1 Introduction

Meta-Optimization is a second-order optimization paradigm in the field of algorithm design (Stephenson et al. 2003; Campbell et al. 2019; Lee et al. 2019). It is centered on automating the design and performance breakthroughs of optimization algorithms themselves by establishing an algorithm generation framework that transcends a specific problem domain (Diakonikolas et al. 2019; Qin et al. 2023; Águila-León et al. 2024). This paradigm has shown revolutionary potential in the fields of complex system optimization, combinatorial decision science, and computational intelligence (Mohammed et al. 2019; Reif et al. 2012; Vlahogianni et al. 2005). It has shown perfect performance, especially in dealing with the limitations of traditional algorithms such as NP-hard problems (Hochba 1997; Son

Extended author information available on the last page of the article

et al. 2024), adaptive dynamic environment (Nguyen et al. 2012; Wang et al. 2024b), and intelligent optimization (Liu et al. 2024f, d). Recent studies identify inherent limitations in traditional metaheuristic algorithms, such as Genetic Algorithms and Particle Swarm Optimization(PSO) (Ma et al. 2025b; Hayward and Engelbrecht 2025; Fu et al. 2025). These limitations involve parameter sensitivity, premature convergence, and deficient cross-domain generalization (Cheng et al. 2025; Li et al. 2025d). Consequently, their effectiveness in industrial-grade complex systems shows significant degradation (Aryan et al. 2025; Siddharth Kashyap et al. 2025). This dilemma highlights the urgent need for fundamental changes in Evolutionary Algorithm(EA) design. Existing methods heavily depend on manual approaches based on experience. Future solutions should emphasize automated and data-driven strategies. These changes are crucial to address the current limitations (Shamim Ahsan et al. 2025; Citterio and Tangherloni 2025; Cheggour and Loscri 2025) (see Fig. 1).

In the rapid development of AI technology, Large Language Models (LLMs), represented by GPT, have become the significant part that drives technological innovation in multiple domains due to their breakthrough semantic understanding and generation capabilities (Brown et al. 2020; McDuff et al. 2025; Yuksekgonul et al. 2025; Dhakal and Parry 2024). These models have completely reconfigured the fundamental paradigm of Natural Language Processing (NLP) (Chang et al. 2024). Taking the medical field as an example, LLMs have been able to parse the medical literature to generate diagnostic suggestions (Thirunavukarasu et al. 2023), or assist clinical decision making through multimodal input, which significantly improves the efficiency and precision of medical services (Li et al. 2025a; Ferber et al. 2024). These advances not only validate the generalization ability of LLMs in complex tasks but also provide a reusable technical paradigm for algorithm design automation.

Current algorithm engineering faces two major structured challenges: first, the trial-and-error-based hyperparameter tuning process, which relies heavily on expert experience, is highly subjective and time-consuming. In modern optimization problems with exponentially increasing hyperparameter space dimensions, this approach is unsustainable (Yang et al. 2025a; Kumar 2022; Rajwar and Deep 2025); second, the theoretical frameworks of existing meta-heuristic algorithms are mostly based on simplifying assumptions, which

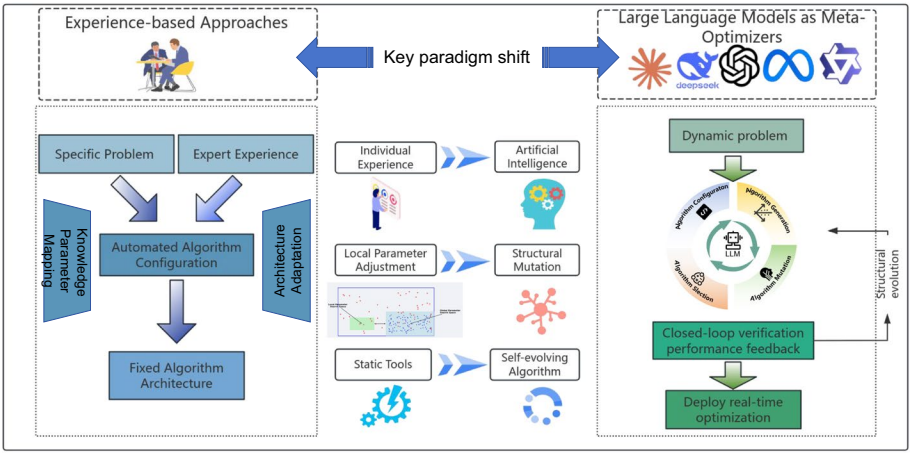


Fig. 1 The differences between traditional experience-based approaches and LLM as meta-optimizers

make it difficult to efficiently capture the nonlinear coupling characteristics and dynamic evolutionary laws of the real-world problems (Fu et al. 2024b; Song et al. 2024). For example, in the power grid configuration optimization scenario, the adaptive error rate of the standard genetic algorithm to demand fluctuations is significantly higher than that of the manually adjusted hybrid strategy (Heng and Rahiman 2025; Micev et al. 2025).

Driven by these limitations, the academic community has been motivated to explore new avenues in Machine Learning (ML) -based Algorithm Generation. Notable progress has been achieved in Neural Architecture Search (NAS) (Lei et al. 2025; Qu et al. 2025) and the design of combinatorial optimizers based on graph networks within the Reinforcement Learning (RL) framework (Inoue et al. 2025; Zhang et al. 2025b). By leveraging the synergistic potential between gradient-free optimization heuristics and generative AI, the integration of LLMs with Evolutionary Computation(EC) has emerged as a transformative frontier (Liu et al. 2023; Xiao et al. 2024; Araujo 2007). This fusion is driving innovation across four key dimensions of algorithm design:

1.Automated Algorithm Synthesis and Adaptation: LLMs transcend manual coding constraints by generating executable optimization algorithms from textual specifications (Tang et al. 2025; Cui et al. 2025). They dynamically optimize the algorithm code workflow through the algorithm evaluation and innovate operators and pipeline architectures in response to problem dynamics (Li et al. 2025b, g). When it comes to algorithmic evolution, compared to manually designed evolutionary operators and evolutionary strategies, LLMs have a significant advantage in discovering novel evolutionary operators and evolutionary strategies for complex and difficult optimization problems.

2.Intelligent Algorithm Selection: Functioning as meta-reasoners, LLMs analyze problem characteristics, including constraint topology and fitness landscape properties, to prescriptively identify optimal evolutionary strategies (Shu and Du 2024). This enables real-time switching between algorithm families during optimization cycles (Wu et al. 2023; Tian et al. 2025).

3.Smart Hyperparameter Configuration: Replacing empirical tuning, LLMs decode performance feedback patterns to autonomously configure evolutionary parameters (Wu et al. 2025; Chen et al. 2024b; Li 2025). By correlating solution quality trajectories with parameter spaces, they establish adaptive control policies for population size, selection pressure, and operator frequencies.

4.Adaptive Mutation Governance: LLMs introduce context-sensitive control over evolutionary stochasticity (Li et al. 2025f; Sen et al. 2025). Through continuous monitoring of diversity metrics and convergence signatures, they probabilistically orchestrate mutation types, intensities, and application targets to balance exploration-exploitation tradeoffs (Hassan et al. 2024; Eom et al. 2024).

For the design of heuristic algorithms and meta-heuristic algorithms that are strongly related to high-quality expert experience, intelligent systems based on LLMs are able to break through the spatial and temporal limitations of traditional expert experience (Tian et al. 2024; Zhao et al. 2025b)and realize paradigm innovation of algorithm design through their powerful knowledge representation and reasoning capabilities (Wei et al. 2025b; Tian et al. 2025).

This survey is structured as follows. Section 2 establishes the foundational background and compares LLMs with traditional methods. Section 3 then delves into the core methodologies of LLM-driven meta-optimization. Subsequently, Sect. 4 examines advanced

enhancements, while Sect. 5 covers evaluation metrics, real-world applications, and benchmark limitations. Finally, Sects. 6 and 7 discuss the prevailing challenges and outline future research directions, respectively. The specific article structure is shown in Fig. 2. And some important abbreviations and their full names are listed in Table 1.

2 Background and related work

This section provides a foundational overview of meta-optimization and LLMs, setting the stage for understanding how LLMs are being used in automated algorithm design. Unlike existing surveys that focus on LLM-as-tool paradigms, our work redefines LLMs as autono-

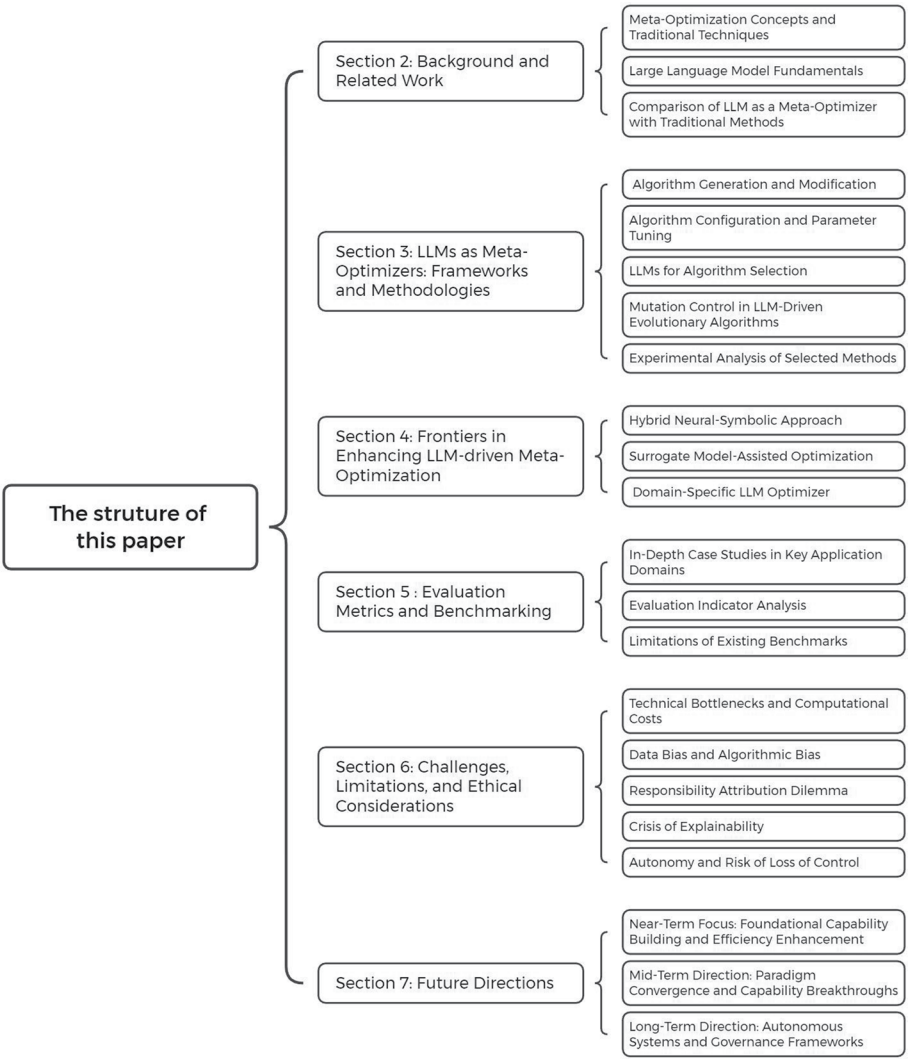


Fig. 2 The structure of this paper

Table 1 Abbreviation

Abbr.	Full Name
LLM	Large language model
NLP	Natural language processing
NAS	Neural architecture search
BBO	Black-box optimization
MetaBBO	Meta-black-box optimization
EA	Evolutionary algorithm
PSO	Particle swarm optimization
CS	Cuckoo search
ML	Machine learning
ABC	Artificial bee colony
RL	Reinforcement learning
AMP	Automated modeling and problem-solving
RAG	Retrieval-augmented generation
CPS	Cyber-physical systems
BO	Bayesian optimization
EC	Evolutionary computation
SOTA	State-of-the-art
TSP	Traveling salesman problem

mous meta-optimizers with closed-loop control capabilities. This fundamental shift enables breakthroughs in semantic-level mutation governance and industrial-grade reliability verification, gaps unaddressed in prior literature. A comparison with similar works is shown in Table 2.

2.1 Meta-optimization concepts and traditional techniques

Meta-optimization, particularly Meta-Black-Box Optimization (MetaBBO), is employed to automatically design Black-Box Optimization (BBO) algorithms, thereby reducing the reliance on expert knowledge and manual trial-and-error efforts. (Lu et al. 2025; Yang et al. 2025c; Shao et al. 2025). In BBO, the objective function and constraints are not explicit functions of the decision variables but are instead provided through some unknown, implicit manner. This means that the analytical expressions or mathematical models of the objective function and constraints cannot be directly obtained, and their values must be obtained through evaluation functions (Golovin et al. 2017; Yang et al. 2025b). Typically, experiments, simulations, or other computational methods are used to evaluate the values of the objective function and constraints. These evaluation functions take the decision variables as inputs and calculate the corresponding objective function and constraint values. By continuously trying different values of the decision variables and performing optimization searches based on the results returned by the evaluation functions, the optimal solution is found. BBO problems are diverse, covering single-objective, multi-objective, constrained, multimodal, large-scale, combinatorial, and mixed-integer linear programming challenges (Li et al. 2025e; Zhu et al. 2025; Hoang et al. 2025; Boyne et al. 2025). MetaBBO is a framework that uses meta-learning technology to automatically design or adapt BBO algorithms (Ma et al. 2025a). It aims to solve core issues in traditional BBO, such as reliance on manually designed rules, limited generalization capabilities, and performance degradation

Table 2 Comparison of different survey works

Methods	AG	AC	AS	IA	AM	EM	S/E	DS	BC
LLM+EA Yu and Liu (2024)	✓	✓				Basic			Standard
Prompt Li (2023)						Basic			NLP tasks
LLM4AD Liu et al. (2024b)	✓	✓	✓	✓		Basic			Standard
MetaBBO Ma et al. (2025b)	✓	✓	✓			Advanced			BBO focus
LLM+EA Roadmap Wu et al. (2024)	✓			✓		Basic	Limised		Standard
LLM+Opt Huang et al. (2024b)	✓	✓		✓		Basic			Dynamic env
This work	✓	✓	✓	✓ (CPS/Power)	✓	Novelty +Safety	✓ (CPS focus)	✓ (Edge-Cloud)	Industrial CPS

AG algorithm generation, *AC* algorithm configuration, *AS* algorithm selection, *Bsc* basic, *IA* industrial application, *AM* algorithm mutation, *EM* evaluation metrics, *Adv* advanced, *S/E* safety & explanibility, *DS* deployment strategies, *Nov* novelty

in high-dimensional scenarios (Yin et al. 2024). It brings new transformations to diverse practical application fields (Tlili et al. 2025; Yun et al. 2024).

Meta-optimization techniques can be broadly categorized, encompassing Evolutionary Algorithms (EAs) (Yu and Gen 2010), Bayesian Optimization (BO) (Shahriari et al. 2015), and Reinforcement Learning (RL) (Kaelbling et al. 1996). EA is inspired by biological evolution and is effective in solving complex optimization problems, especially those that are large-scale, high-dimension, and dynamically changing (Bäck and Schwefel 1993; Fortin et al. 2012). However, traditional EA often necessitate carefully designed operators and domain expertise to achieve satisfactory performance, limiting their adaptability to new and unseen complex optimization problems (do Val Lopes and Machado 2025; Asadi et al. 2025; He et al. 2025).The swarm intelligence algorithms based on this are generally hybrid types, such as PSO (Kennedy and Eberhart 1995), Cuckoo Search (CS) (Yang and Deb 2009), and Artificial Bee Colony (ABC) (Karaboga et al. 2005), also play a significant role. Bayesian Optimization, while not explicitly detailed in terms of algorithm flow in the provided digests, is listed as a classic BBO algorithm (Pelikan and Pelikan 2005). RL is implicitly referenced as a meta-optimization technique, as it reduces the need for expert knowledge (Zhang et al. 2022; Gupta et al. 2018). However, traditional approaches have limitations, including reliance on expert knowledge, high computational costs, and challenges in the transfer of knowledge between different domains of problem. Traditional Automated Modeling and Problem-solving (AMP) systems can be "open-box," requiring detailed business logic, or "closed-box," using neural network approximations, each with their own drawbacks (Ohsuga 2022; Rojas et al. 2023). These limitations of traditional meta-optimization

techniques highlight the need for a new paradigm that can reduce reliance on expert knowledge and improve generalization. The recent emergence of LLMs offers a promising pathway to address these challenges, which we will introduce next.

2.2 Large language model fundamentals

LLMs have revolutionized various fields, and their emergence is closely tied to the development of the Transformer architecture, which uses attention mechanisms to capture long-range dependencies in text (Vaswani et al. 2017; Han et al. 2021). The training process involves pre-training on vast datasets, followed by fine-tuning for specific tasks (Zhao et al. 2023b). Studies have shown that model performance follows the Scaling Laws, i.e., the increase in parameter size, data volume, and computational resources is power-law related to the increase in capability, but this process is accompanied by diminishing returns and unpredictable phase transitions (Kaplan et al. 2020; Aghajanyan et al. 2023). Current mainstream LLMs further extend their applicability in multilingual and cross-domain tasks through technological innovations such as Mixture-of- Experts, MoE architecture (Rajbhandari et al. 2022; Lin et al. 2024), and Retrieval-Augmented Generation (RAG) (Lewis et al. 2020; Gao et al. 2023). LLMs possess key capabilities such as natural language understanding and generation, code generation, and reasoning (Huang and Chang 2022; Shi et al. 2023). Prompt engineering, including techniques like "chain of thought" prompting, is crucial for guiding LLMs to (Wei et al. 2022; Lyu et al. 2023) perform complex tasks effectively (Wei et al. 2022; Lyu et al. 2023).

Different types of LLMs offer varying architectures, training data, and performance characteristics. Techniques like prompt engineering, in-context learning, and fine-tuning further enhance their performance. A comparison of closed-source models like GPT-4o (Achiam et al. 2023) and open-source models like DeepSeek (Guo et al. 2025) reveals trade-offs in terms of performance and accessibility.

2.3 Comparison of LLM as a meta-optimizer with traditional methods

Before establishing LLMs as emerging meta-optimizers, the field of automated algorithm design was long dominated by traditional computational intelligence methods (Zhao et al. 2025a). To accurately evaluate the contributions and applicability of both LLMs and traditional methods, a clear understanding of their relationship is required. Furthermore, a direct comparison of their strengths and weaknesses is essential.Traditional intelligent optimization algorithms rely on mathematical frameworks or bio-inspired search and are known for their high computational efficiency on specific problems (Zong and Wan 2025). Genetic Programming (GP) evolves algorithmic structures from a component library, providing both powerful search and white-box transparency (Zhong et al. 2025). However, it relies on carefully designed function sets, lacks semantic understanding, often generates redundant code, and exhibits limited generalization. BO uses probabilistic surrogate models to find near-optimal solutions with minimal evaluations, featuring low computational overhead (Shahriari et al. 2015). Yet, it is constrained by model accuracy, suffers performance degradation in high dimensions, and struggles with algorithmic logic generation. A shared advantage of these methods is their minimal reliance on large-scale GPU clusters, with computational overhead primarily stemming from objective function evaluation, making them suitable for

resource-constrained environments. The introduction of LLMs represents a paradigm shift, with their core advantage rooted intheir superior semantic understanding and generative capabilities. As shown as Fig. 3. LLMs do not replace traditional methods but complement them: they hold immense potential in scenarios requiring rapid prototyping, semantic understanding, and strategy innovation; whereas in resource-constrained environments demanding high reliability and interpretability, GP and BO remain more robust and economical choices. Future research should focus on integrating LLM semantic reasoning with the computational efficiency of traditional methods to build hybrid meta-optimizers, jointly advancing the field.

This comparative analysis underscores the unique value proposition of LLMs as meta-optimizers. In the following section, we will delve into the specific frameworks and methodologies that realize this potential, detailing how LLMs are employed for tasks such as algorithm generation, configuration, and mutation control.

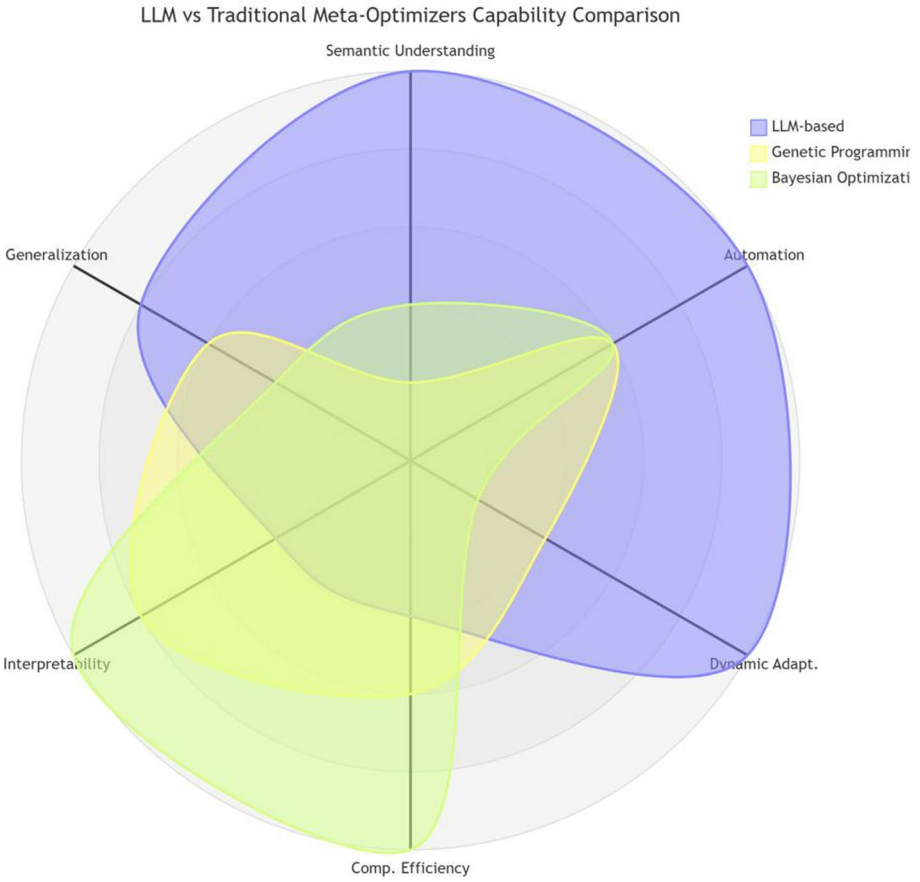


Fig. 3 Comparison of LLM as a meta-optimizer with traditional methods

3 LLMs as meta-optimizers: frameworks and methodologies

This section aims to delve into existing frameworks and representative methods that utilize LLMs as meta-optimizers to drive the automated design of intelligent optimization algorithms. We will systematically analyze the specific applications of LLMs across key dimensions of automated algorithm design, progressing from high-level concepts to concrete operational paradigms.The core idea is to leverage the capabilities of LLMs in understanding, generating, and manipulating code and mathematical models to enhance and automate various aspects of algorithm creation, configuration, and selection (Naveed et al. 2023). This also includes using LLMs to convert natural language descriptions of optimization problems into mathematical programming models and iteratively refine these models using prompt optimization techniques (Huang et al. 2023). The following subsections explore specific applications of LLMs in automated intelligent optimization algorithm design. "LLMs for Algorithm Generation and Modification" examines how LLMs can generate entire algorithms or modify existing ones, with a focus on meta-search algorithms and their components. "LLMs for Algorithm Configuration and Parameter Tuning" discusses how LLMs can be used for hyperparameter optimization and algorithm configuration, employing techniques such as iterative prompt optimization and parameter-efficient fine-tuning. 'LLMs for Algorithm Selection' investigates how LLMs can automatically select the most suitable algorithm for a given problem considering factors such as the characteristics of the problem and the consumption of resources. Finally, 'Mutant Control in LLM-Driven EAs' delves into how LLMs can dynamically adjust mutation strategies in EAs to balance exploration and exploitation, thus enhancing the adaptability and convergence of the algorithm. A comprehensive overview of the relevant research methods is presented in Table 3.

3.1 LLMs for algorithm generation and modification

LLMs are increasingly being used to generate novel algorithms or modify existing ones, offering a new paradigm in automated algorithm design (van Stein et al. 2024; Zhao et al. 2023a). This capability stems from LLMs' ability to understand and generate code, making them suitable for creating algorithms from scratch or refining existing ones. The mainstream framework centers on semantic understanding and dynamic evolution, the system first defines natural language descriptions of problems and initial algorithm templates, then converts them into a machine-readable optimization space through constraint coding. LLMs act as generation engines, driving parallel algorithm construction and component-level modification within this space. Real-time benchmarking feedback and performance evaluation reports are obtained through prompt engineering-guided evolutionary iteration, continuously optimizing algorithm structure and functional modules. The paradigm of LLMs for algorithm generation and modification is shown in Fig. 4.

Several landmark studies have already demonstrated how LLMs can assist in algorithm generation.Here, we first present two representative approaches Funsearch (Romera-Paredes et al. 2024) and EoH (Liu et al. 2024a).

The FunSearch method (Romera-Paredes et al. 2024), for instance, embeds a frozen, pretrained code-LLM inside a distributed island-model genetic framework. It searches directly in function space. A problem is specified as an executable and score-able evaluate function together with a human-written skeleton. Only the parts of the skeleton explicitly marked

Table 3 A concise map of LLM–evolutionary algorithm synergy

Category	Method	Role of LLM	Prompt strategy	Benchmark/task
AG	van Stein and Bäck (2024)	LLMaD	CoT	BBOB Suite
	Lu et al. (2024)	LLMaO	FS	TMC Optimization
	Surina et al. (2025)	LLMaD	FS	Bin Packing, TSP, FlatPack
	Liu et al. (2024a)	LLMaD	FS+CoT	TSP, Online-BPP, FSSP
	Romera-Paredes et al. (2024)	LLMaD	FS	Cap Set, Online-BPP
	Liu et al. (2024e)	LLMaO	FS	MOEA Benchmarks
	Ali et al. (2025)	LLMaS	CoT	TSP
AC	Shi et al. (2025)	LLMaO	ZS	Multi-domain LLM Training
	Salimian et al. (2025)	LLMaO	FS	Text Classification
	Bosio and Mueller (2024)	LLMaD	FS	Pendulum, Ball-in-Cup
	Kong et al. (2025)	LLMaO	FS	PAIR, Psych8k
AS	Wu et al. (2023)	LLMaE	ZS	AS Benchmarks
	Videau et al. (2024)	LLMaE	FS	GSM8k, MathQA
AM	Zhang et al. (2025a)	LLMaO	FS	7 Multi-Agent Benchmarks
	Zhang et al. (2024a)	LLMaO	FS	47 General Tasks
	Lee et al. (2025a)	LLMaO	CoT+FS	TravelPlanner, Natural Plan
	Cui et al. (2024)	LLMaO	FS	35 Benchmark Tasks

AG algorithm generation, *AC* algorithm configuration, *AS* algorithm selection, *AM* algorithm mutation, *ZS* zero-shot, *FS* few-shot, *CoT* chain of thought

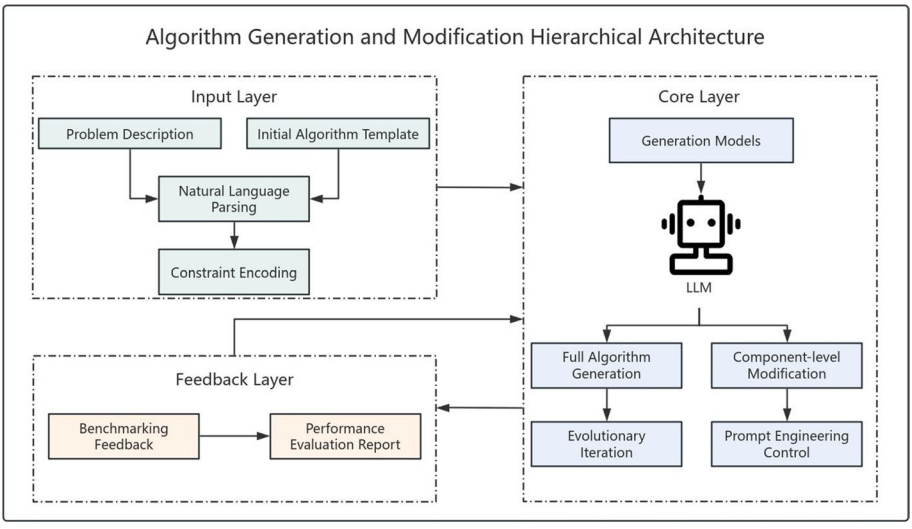


Fig. 4 The paradigm of LLMs for algorithm generation and modification

as mutable are evolved. The system keeps several parallel "islands". Each island stores the highest-scoring, semantically correct function variants seen so far. When an island needs a new individual, it first samples a high-fitness cluster via a Boltzmann mechanism, then re-samples within that cluster, favouring shorter programs. Two representative functions and the skeleton are assembled into a prompt. The LLM completes a new function body in a zero-shot manner. The new function is executed and scored by an asynchronous distributed

evaluator. Valid code is written back to the corresponding island. Every four hours, a global shuffle occurs: the worst half of the islands are discarded, and the best individuals from the survivors are used to re-seed them. This preserves diversity while pushing the population toward globally high-fitness regions. The main task logic is shown in the Fig. 5.

EoH (Liu et al. 2024a) couples a LLMs with evolutionary computation to automate heuristic design with minimal prior knowledge. Each heuristic is represented in two modalities: a natural-language "thought" that captures the high-level idea, and executable "code" that realises it. These two layers act as a single chromosome and co-evolve within one loop, jointly exploring conceptual novelty and implementation details, thus mirroring the human practice of "think first, code later".The algorithm keeps a fixed-size population. In every generation, five carefully designed prompting operators, E1, E2, M1, M2, M3, turn the LLM into both mutation and crossover engines. E1 urges the model to depart radically from parents at the thought level; E2 extracts a shared skeleton from multiple parent thoughts and elaborates a new idea upon it; M1–M3 locally refine a single thought or code, tune parameters, and trim redundancy. Each freshly generated thought–code pair is compiled on the fly and scored on representative instances. Survivors enter the population, and truncation selection retains the elite. EoH requires no domain priors or hand-crafted parts. The main task logic is shown in the Fig. 6, and the core function prompt is also shown in the Fig. 7.

Current research on algorithm generation and modification has formed two major interdisciplinary paradigms: LLM-driven meta-algorithm generation and component-level semantic reconstruction. At the meta-algorithm level, frameworks such as LLaMEA (van Stein and Bäck 2024) utilize GPT models to directly generate the complete code architecture of meta-heuristic algorithms and iteratively optimize the generated fragments through an evolutionary process; Similarly, LLM-EO (Lu et al. 2024) guides multi-objective algorithm design through natural language instructions, while EvoTune (Surina et al. 2025) constructs a generation-evaluation-RL feedback loop to achieve dynamic strategy updates. Component-level reconstruction focuses on semantic transformations of algorithm core

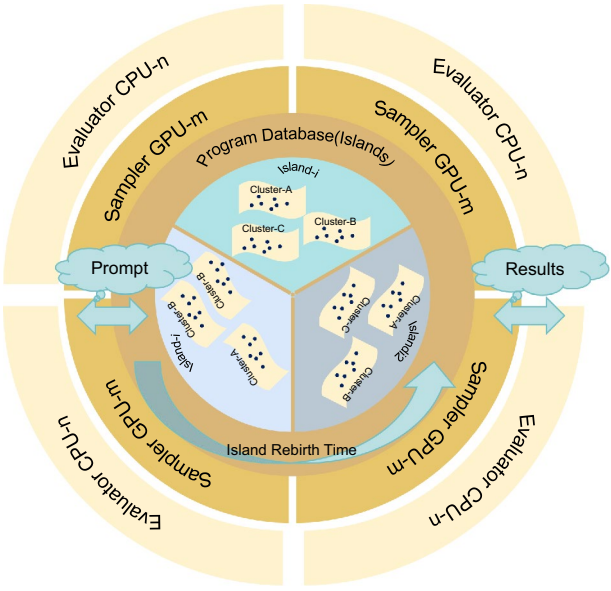


Fig. 5 The structural diagram of FunSearch

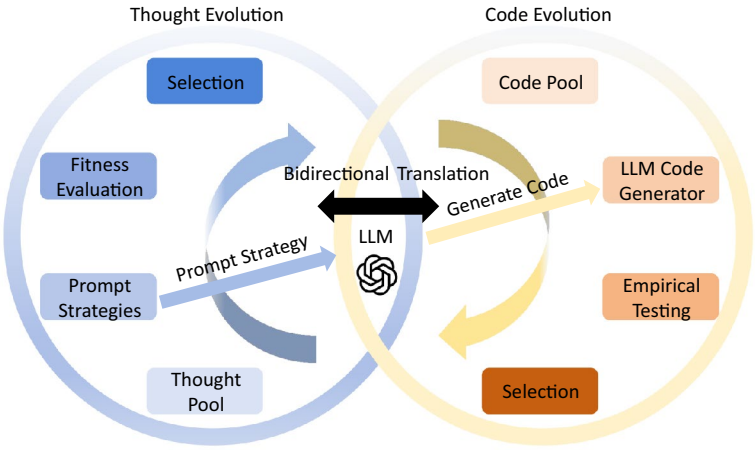


Fig. 6 The structural diagram of EoH

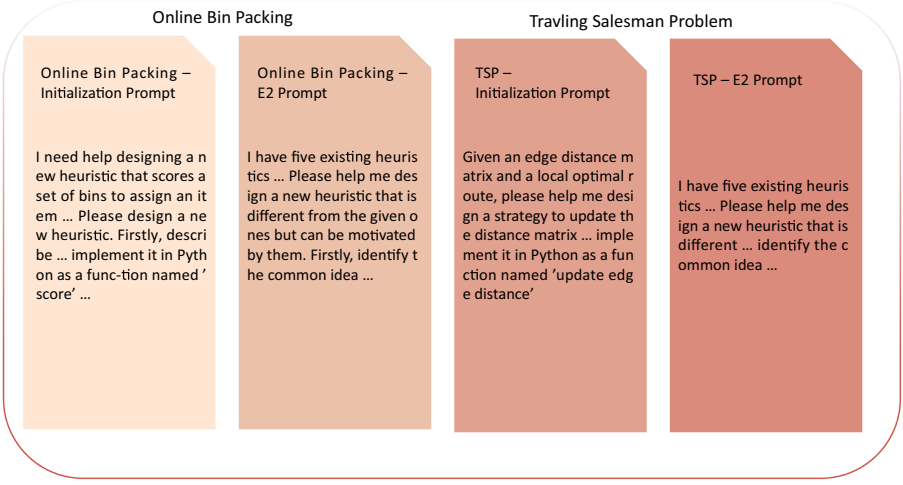


Fig. 7 The main prompt of EoH

operators: MOEA cross-mutation operator generation research (Liu et al. 2024e) guides LLM design of specific operators through information-intensive prompting techniques, PAIR (Ali et al. 2025) uses LLMs as intelligent pairing engines for EAs, restructuring selection logic; hybrid algorithm generation work (Surendar 2025) combines existing algorithm modules through structured prompts.

Prompt engineering serves as the key hub connecting the two paradigms: the OPRO framework (Zhang et al. 2024b) iteratively optimizes meta-prompts based on performance scores, driving directional improvements in algorithm generation; ChainBuddy (Zhang and Arawjo 2024) generates scalable algorithm pipeline drafts through conversational interaction, highlighting the design potential of human–machine collaboration. Quality assessment typically relies on validating performance against benchmark problems, yet faces the dual

challenge of ensuring originality while balancing exploration and exploitation. ReEvo (Ye et al. 2024)combines evolutionary algorithms with LLMs' reflection: evolutionary search efficiently explores the heuristic space, while LLMs provide directional guidance by analyzing performance discrepancies. This approach achieves outstanding results across multiple combinatorial optimization problems, balancing search efficiency with exploration capabilities.

Despite demonstrating formidable semantic comprehension and code generation capabilities in algorithm generation and modification, LLMs still face certain limitations in practical application. Their advantages primarily manifest in two aspects: first, they can transcend the constraints of traditional expert knowledge to generate structurally novel algorithmic variants; second, they significantly lower the barrier to algorithm design through natural language interaction. However, LLMs also exhibit notable shortcomings. The algorithms they generate often lack interpretability, and their black-box nature complicates verification. Additionally, they are highly sensitive to the quality of initial prompts and examples, leading to unstable output. Furthermore, computational overhead is substantial, particularly pronounced in distributed evaluation frameworks. During generation, an inherent trade-off between diversity and reliability must be addressed: excessive pursuit of novelty may render algorithmic behavior uncontrollable, while imposing too many constraints may stifle the creativity of LLMs.

3.2 LLMs for algorithm configuration and parameter tuning

The rise of LLMs is driving a paradigm shift in algorithm configuration toward automation and intelligence. When using LLMs for parameter configuration and tuning, the traditional optimization path is typically restructured through a hierarchical collaborative architecture. This involves first integrating optimization objectives, resource constraints, and parameter models to define foundational constraints. Subsequently, a feedback mechanism is established based on performance evaluations to dynamically analyze multi-dimensional metrics. The LLMs serve as the intelligent hubs, enabling adaptive model configuration and iterative optimization strategies to compress and fine-tune the parameter space. The output layer generates the optimal solution set, the adaptation layer relies on hardware collaboration modules and dynamic controller balancing algorithms to explore behavioral patterns. The paradigm of LLMs for algorithm configuration and parameter tuning is shown in Fig. 8.

LLM-GE (Morris et al. 2024) embeds LLMs directly into the three canonical operators of EAs, genome representation, mutation, and recombination, thereby recasting NAS as an autonomous, self-reflective program-synthesis process steered by the LLM itself. The key idea is to decompose a SOTA network into interpretable Python classes. Instead of acting merely as a hyper-parameter recommender, the LLM now functions as the actual mutation and crossover engine. To counteract the LLM's tendency toward mode collapse and hallucination in code generation, the authors devise a two-tier prompting strategy. In the first tier, LLM-GE assigns distinct personas, such as "top expert," "eccentric scientist," and others, to the LLM and pair them with six meticulously hand-crafted prompt templates: routine or unusual hyper-parameters, complex structures, pruning strategies, unconventional enhancements, and large-scale refactoring. Together, these yield 24 heterogeneous prompt combinations that substantially expand the latent search space. The second tier imports the spirit of Auto-CoT into the evolutionary loop: after each generation, elite individuals identified by

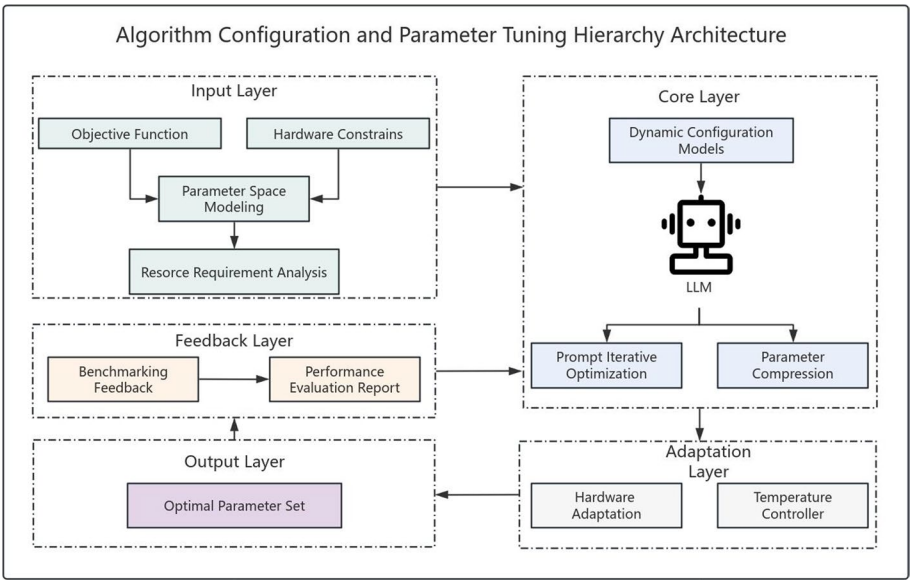


Fig. 8 The paradigm of LLMs for algorithm configuration and parameter tuning

SPEA-2 (Zitzler et al. 2001) are fed back into the LLMs, which is asked to articulate, step-by-step, why the mutation improved performance, SPEA-2 was adopted to leverage its powerful external elite archive mechanism, which ensures that high-performance model variants are not lost in the vast and unstable search space generated by LLMs, and effectively maintains the diversity of solutions. It then generates new, more targeted improvement prompts, enabling vertical knowledge transfer and cumulative refinement across generations. The main task logic is shown in the Fig. 9, and the core function prompt is also shown in the Fig. 10.

The methods of LLM-assisted algorithm configuration and parameter tunning can be categorized into two core approaches: the first is dynamic feedback-based configuration, with representative examples such as the DIDS framework (Shi et al. 2025), which uses gradient clustering and Fisher information matrices to build a closed-loop system for dynamically optimizing sampling probabilities in multi-domain training; the PCS framework (Salimian et al. 2025), on the other hand, leverages metamorphosis relationships to generate confidence metrics, combining perceptual differential evolution algorithms to adaptively configure classification thresholds and weights, significantly enhancing robustness in zero-shot tasks. The second is structured static optimization. For example, in the field of program synthesis (Bosio and Mueller 2024), by constraining the temperature coefficient and probability clipping space of LLM, discrete code generation is transformed into controllable symbolic optimization. The EMORL framework (Kong et al. 2025) adopts a hierarchical grid search strategy to compress the hidden state weight search space of multi-objective RL with superlinear efficiency. These methods collectively highlight the multi-level regulatory nature of algorithm configuration, from low-level hyperparameter fine-tuning, mid-level weight configuration, to high-level strategy generation, gradually forming a "feedback-structure-language" three-in-one methodological framework. Future research should focus

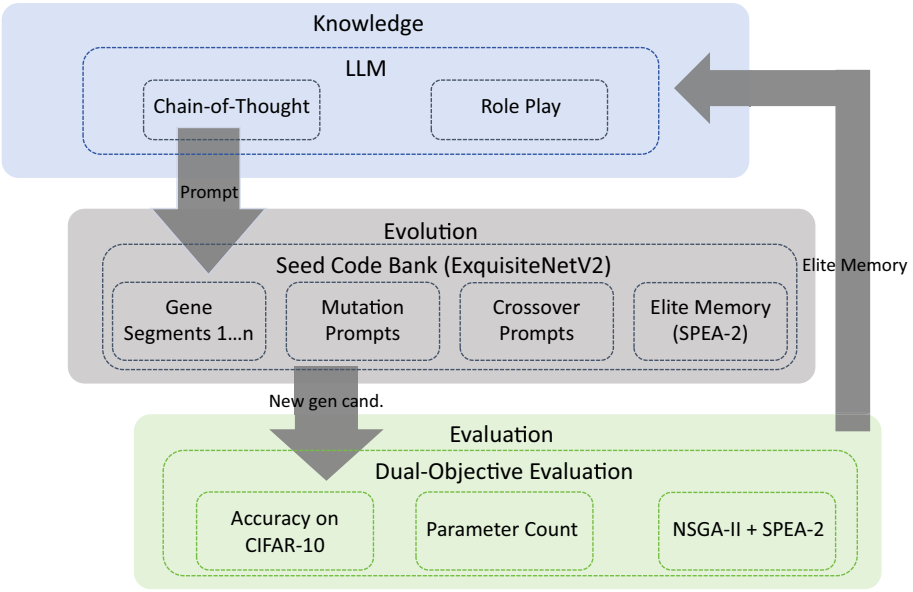


Fig. 9 The structural diagram of LLM-GE

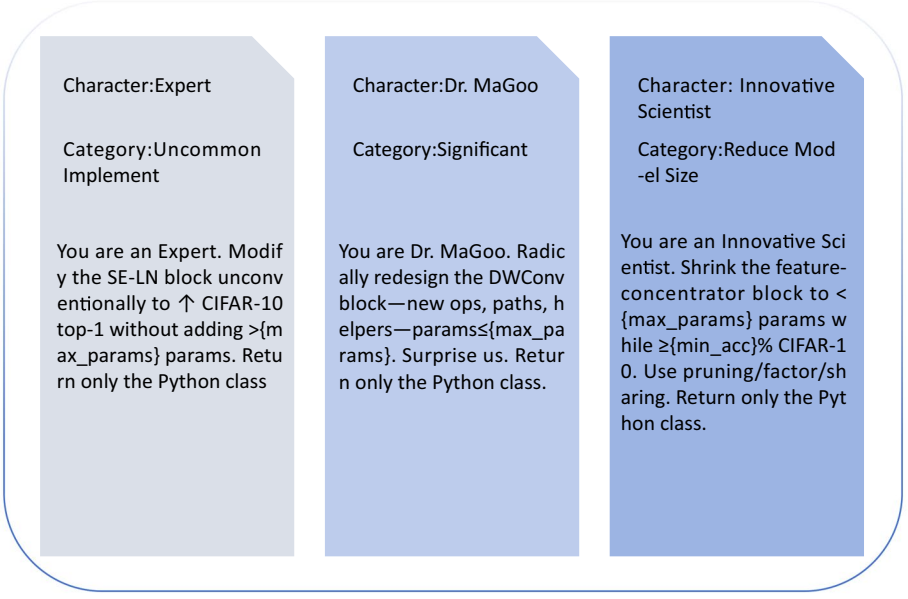


Fig. 10 The main prompt of LLM-GE

on integrating dynamic response mechanisms with LLMs' generalization guidance capabilities, particularly in establishing cross-paradigm collaborative theories in hyperparameter optimization and multi-objective trade-off scenarios, to address the inherent bottlenecks of traditional methods in computational efficiency, environmental adaptability, and human–machine collaboration.

The core advantage of LLMs in parameter tuning lie in its ability to extract abstract patterns from high-dimensional feedback, enabling configuration transfer across problems. However, its primary disadvantage is its high dependency on the training data distribution, resulting in limited generalization capabilities in unknown parameter spaces. Simultaneously, real-time dynamic tuning demands extremely high computational resources, making deployment on edge devices or real-time systems challenging. The critical trade-off lies in balancing automation with interpretability, fully automated tuning, while efficient, sacrifices the controllability and debugability afforded by human intervention.

3.3 LLMs for algorithm selection

LLMs are powerful pattern recognition and reasoning tools that are being actively explored for their potential application in automated algorithm selection processes, with the goal of efficiently identifying optimal solutions from a set of candidate algorithms. This exploration has given rise to a variety of innovative approaches for integrating LLMs into algorithm selection frameworks, forming a diverse set of structured methodologies. The LLM-assisted algorithm selection paradigm manifests as a multi-stage decision flow. The system first integrates problem feature extraction, algorithm characteristic databases, and resource constraints to construct dynamic decision primitives. The LLM acts as a cognitive engine, driving a dual-track mechanism of feature matching and dynamic routing through meta-optimization strategies to coordinate algorithm selection paths. Continuous cost monitoring and performance tracking form a verification loop, continuously calibrating the output of the optimal algorithm and its resource allocation plan. The paradigm of LLMs for algorithm selection is shown in Fig. 11.

AS-LLM (Wu et al. 2023) is a novel framework that, for the first time, injects computable features of the algorithm itself into AutoML algorithm selection, breaking the traditional one-way mapping that relies solely on problem features and past performance.The framework uses a pre-trained LLM to extract deep semantics from source code and generate a high-dimensional algorithm representation, while the objective function of a continuous optimisation problem is traversed as a tree and embedded into a low-dimensional problem representation. Two encoders operate in parallel: an LSTM (Hochreiter and Schmidhuber 1997) encodes the tree-walk sequence for the problem side, and the LLMs, either frozen or fine-tuned, encodes the code text for the algorithm side. Both outputs are pooled and compressed by a two-layer MLP (Hornik et al. 1989) to map them into a shared semantic space. Cosine similarity measures the problem–algorithm compatibility, and a shallow MLP outputs the final selection probability. The main task logic is shown in Fig. 12, and the core function prompt is shown in Fig. 13.

Current algorithm selection methods have established a relatively mature structured paradigm to balance LLMs capabilities with cost-effectiveness, primarily encompassing two categories:

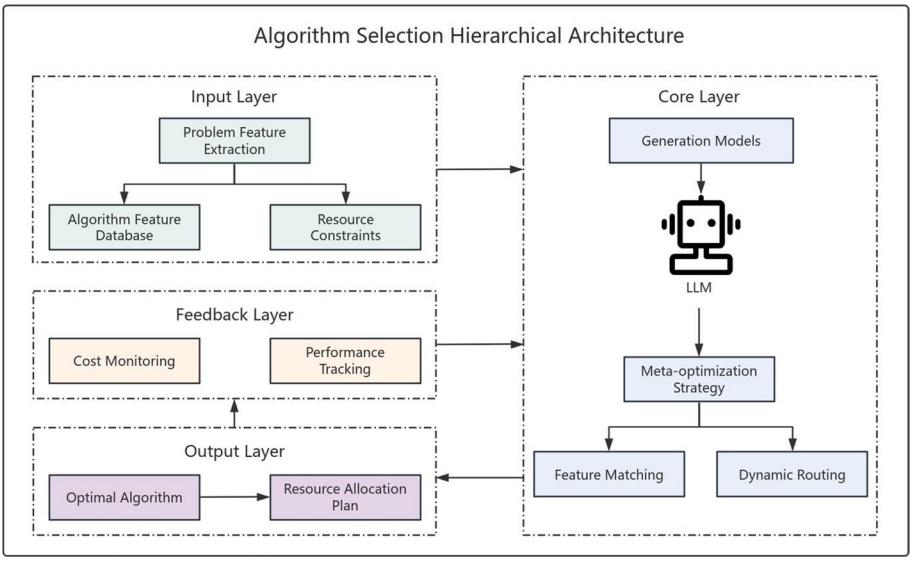


Fig. 11 The paradigm of LLMs for algorithm selection

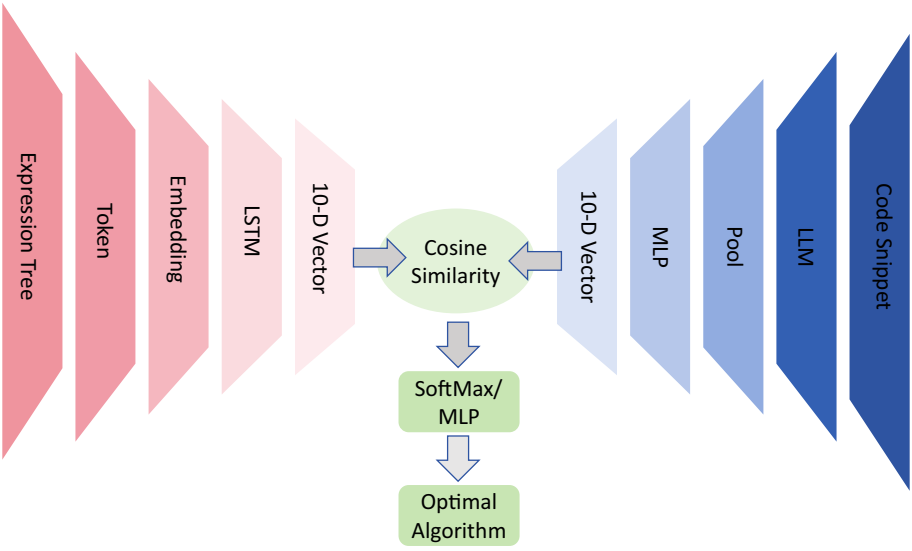


Fig. 12 The structural diagram of AS-LLM

Evolutionary optimization-driven prompt engineering. This category focuses on optimizing prompt combinations as discrete search tasks. The EPPO framework (Videau et al. 2024), for example, systematically explores the performance boundaries of different EAs under information constraints and utilizes comparative feedback mechanisms to balance generalization and optimization performance.

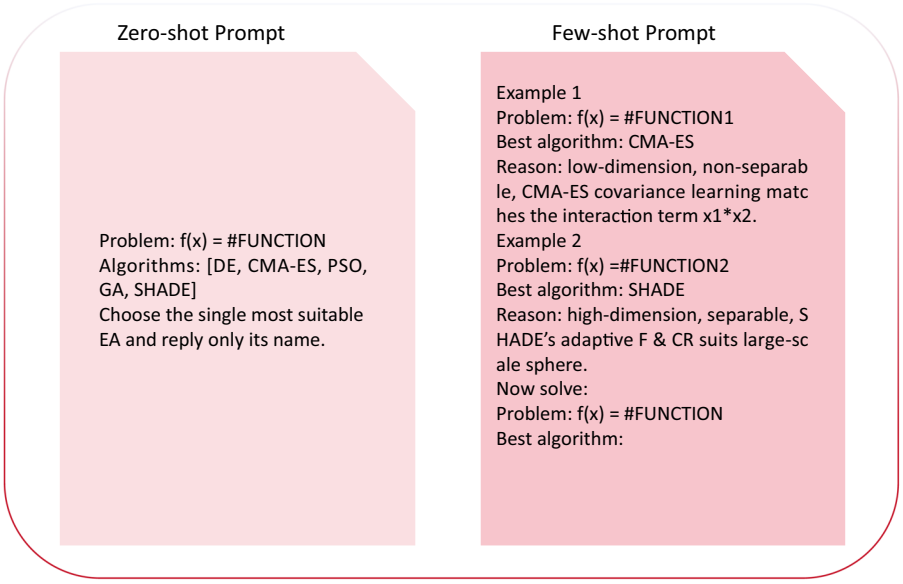


Fig. 13 The main prompt of AS-LLM

Comparison and selection of model enhanced with feature engineering. In complex pattern recognition tasks, these techniques construct high-dimensional statistical features to enhance input representation. Efficient classifiers (Ke et al. 2017) are then selected based on efficiency-generalization trade-offs, often combined with dynamic evaluation mechanisms to achieve scalable selection.

However, the effectiveness of LLM-driven algorithm selection still requires in-depth verification. Its advantages and disadvantages relative to traditional methods are influenced by multiple factors: feature quality and relevance, generalization ability for unseen problems, and its own computational overhead. Clarifying the role of LLMs in dynamic strategies, defining their performance boundaries, and promoting the deep integration of the above structured methods with LLM capabilities are key directions for future development. These methods collectively form the technical pillars for the evolution from static configuration to dynamic routing, from single-model comparison to heterogeneous model collaboration, and from raw input dependency to feature engineering enhancement.

The advantages of LLMs in algorithm selection lie in its robust feature fusion and semantic matching capabilities, enabling it to process multi-source heterogeneous information. However, this approach still has limitations. On one hand, its performance heavily depends on the quality of feature encoding for the problem, and the feature construction process remains largely reliant on manual design. On the other hand, in zero-shot or few-shot scenarios, the model is prone to significant selection bias. The inherent trade-off lies in the conflict between selection accuracy and computational overhead, higher selection accuracy typically comes with greater inference latency, making it difficult to apply to decision-making scenarios with high real-time requirements.

3.4 Mutation control in LLM-driven evolutionary algorithms

Mutation control is a critical aspect of LLM-driven EAs, as it influences the algorithm's ability to effectively explore the search space, adapt to problem characteristics, and avoid premature convergence (Fan and Yan 2015). To address this challenge, several methods have emerged that leverage the capabilities of LLMs to dynamically adjust mutation strategies. LLMs assist algorithm mutation is built on an environment-responsive collaborative evolution mechanism. The system continuously monitors the population gene pool and environmental status, dynamically analyzing evolutionary trends through diversity analysis and convergence detection. LLMs serve as adaptive engines, integrating performance evaluation feedback and optimization strategies to drive dual-scale interventions at the code level and parameter level, ultimately outputting an updated population validated over multiple generations. The paradigm of mutation control in LLM-driven EAs is shown in Fig. 14.

Mind Evolution (Lee et al. 2025a), an inference-time compute strategy that tightly integrates evolutionary search with LLMs to overcome the challenges of complex constraints and vast search spaces in natural-language planning tasks. The method treats the LLMs as a parallel "evolution engine." Candidate solutions are expressed in plain language and refined across generations by the three canonical genetic operators: selection, crossover, and mutation.Mind Evolution adopts an island model to maintain diversity. Each island evolves its own set of dialogue threads. After the LLMs produce an initial plan, a two-turn dialogue acts as local refinement: a critic pinpoints flaws, and an author revises the plan accordingly. Periodic inter-island migration and global restarts inject elite genes while preventing premature convergence. Selection employs a Boltzmann tournament that balances fitness and diversity. During recombination, the LLMs first critique the weaknesses of several parent plans, then synthesises a child plan that merges their strengths. This realises a joint search that is both broad and deep. No formal modelling is required: a programmable evalua-

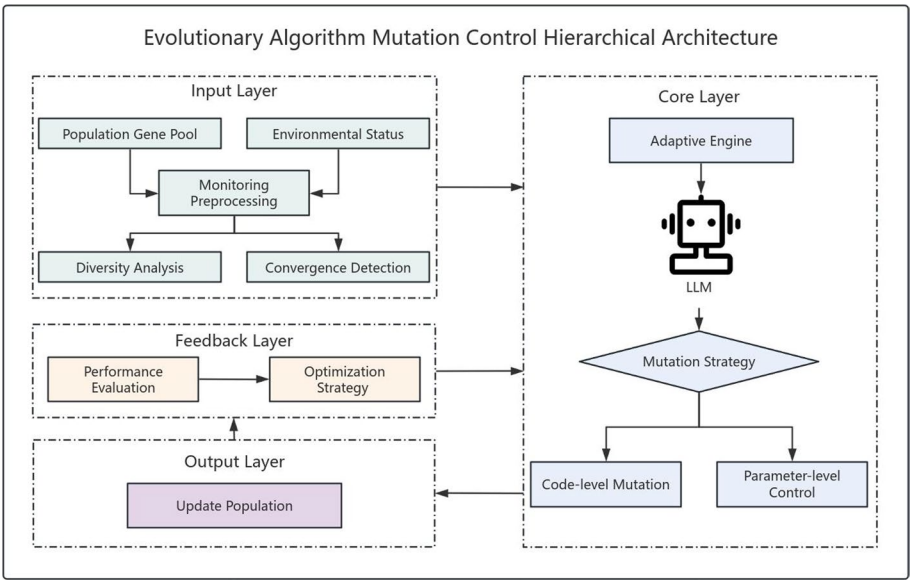


Fig. 14 The paradigm of mutation control in LLM-driven evolutionary algorithms

tor supplies only numerical scores and textual feedback, keeping expert effort minimal. The schematic depicts a ring of islands. In each cycle, the LLMs generate and improve candidate travel plans in parallel; the evaluator scores them and returns natural-language critiques. High-fitness individuals migrate across islands; low-quality islands are reseeded from global elites. Over successive generations, the population converges on a plan that satisfies all constraints, exemplifying an alternating "divergence–convergence" cognitive search pattern. The main task logic is shown in the Fig. 15, and the core function prompt is also shown in the Fig. 16.

LLM-based program semantic mutation mechanism: These methods innovatively use pre-trained LLMs as mutation operators for EAs, directly generating control strategies in the form of Python code. Their core lies in leveraging the LLM's token sampling capabilities for semantic-level program variation: by adjusting temperature parameters to control structural randomness, combined with diversity constraint techniques, they guide directed exploration of the program space. The EvoFlow framework (Zhang et al. 2025a) typically

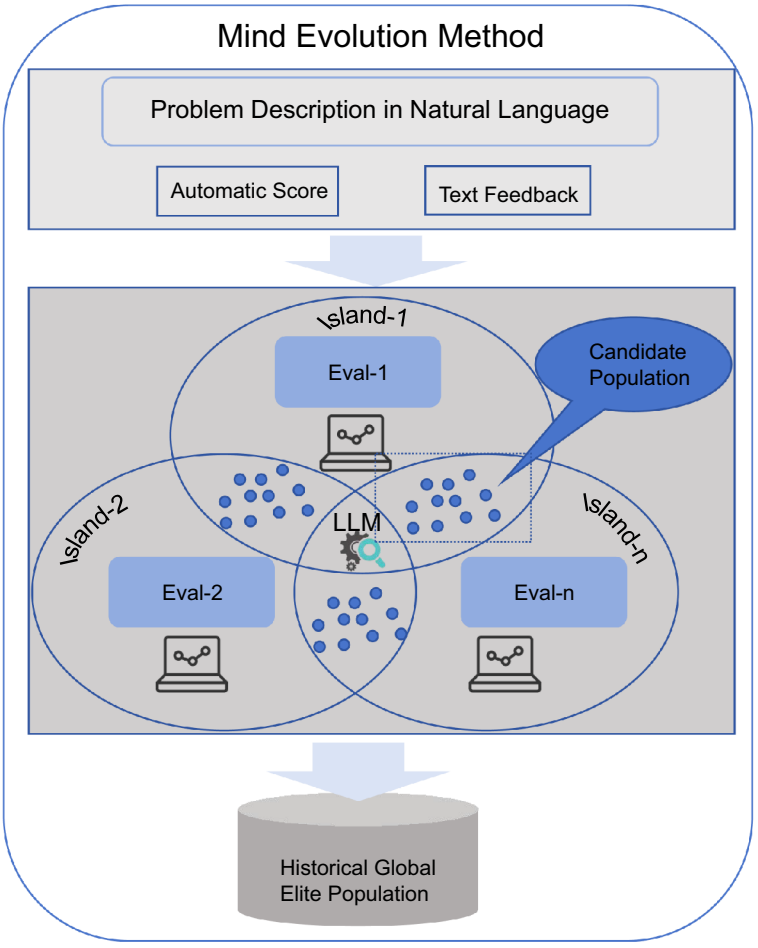


Fig. 15 The structural diagram of mind evolution

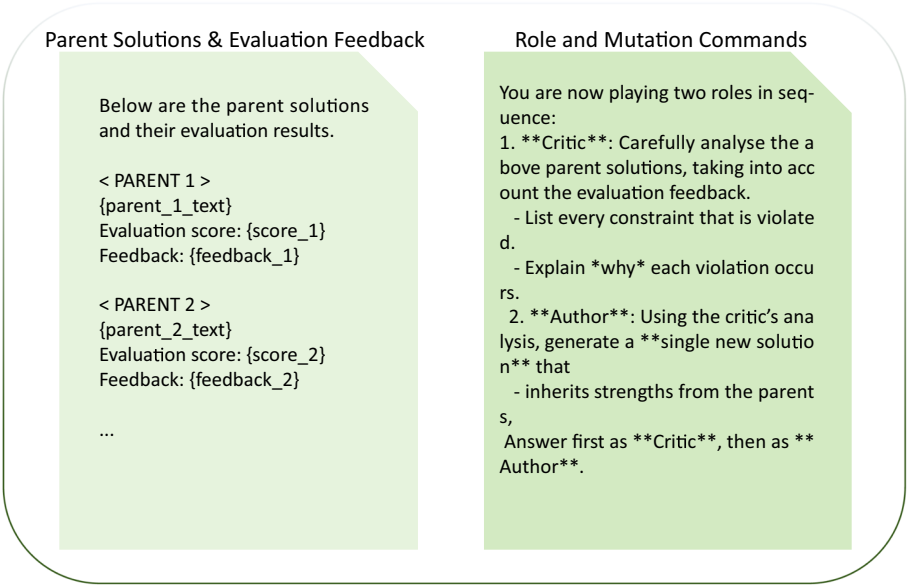


Fig. 16 The main prompt of mind evolution

runs in a parallel architecture, where the LLM receives historical high-performance strategies as contextual prompts to generate new syntactically valid and functionally evolved code variants (Zhang et al. 2024a). The key advantage of this mechanism lies in its implicit semantic mutation capability, which breaks through the limitations of traditional symbolic genetic programming to generate compact, well-formed, and functionally effective code strategies. Importantly, the generated code retains human editability, allowing researchers to intuitively adjust parameters or add logical branches, forming an efficient human–machine collaboration 'mutation fine-tuning' closed loop, providing a new approach to develop explainable and verifiable control strategies for safety-critical systems.

Structured multi-stage mutation framework for prompt optimization: This research focuses on the evolutionary optimization process of prompt engineering, with its innovation lying in the design of specialized multi-stage LLM mutation operators and the construction of an adaptive multi-stage evolutionary pipeline. The PhaseEvo framework (Cui et al. 2024) abandons traditional semantic similarity metrics and instead adopts a diversity assessment method based on performance vectors, significantly enhancing the diversity of mutation generation. Its core lies in designing a task-aware multi-stage pipeline that dynamically configures different mutation operator sequences. This structured "evolution-correction" collaborative strategy elevates prompt optimization from a fragmented operator stacking process to a systematic one, significantly reducing computational resource consumption while greatly enhancing efficiency in multi-objective optimization.

LLM-driven mutation control endows algorithms with enhanced environmental adaptability and semantic perception capabilities. However, its application still has significant shortcomings. On one hand, mutation operations inherently possess unpredictability, potentially disrupting or even undermining the population's convergence process. On the other hand, the method's performance heavily relies on the quality of LLMs reasoning, result-

ing in overall algorithmic instability. Within this mechanism, the critical trade-off lies in the dynamic equilibrium between exploration and exploitation, while LLMs can effectively enhance search diversity, they may also introduce substantial ineffective perturbations that instead delay algorithmic convergence.

3.5 Experimental analysis of selected methods

To demonstrate the capabilities of LLM as a meta-optimizer, we compiled empirical analyses of leading methods on the Traveling Salesman Problem (TSP) (Sui et al. 2025). TSP is a classic NP-hard problem in combinatorial optimization, whose core challenge is finding the minimum-weight Hamiltonian cycle in a weighted graph. As shown in Table 4, the heuristic algorithms generated by the methods discussed in this review have achieved performance levels that match or surpass those of long-established traditional heuristics. This demonstrates that LLM-driven meta-optimization not only replicates human expertise but also possesses the potential to discover innovative strategies that surpass existing designs.

First, traditional heuristic methods demonstrate inherent stability, maintaining relatively consistent performance across varying problem scales. This stems from their deterministic decision rules grounded in human expert experience. However, this stability also implies a clear upper bound on their capabilities. They cannot autonomously learn or adapt from data or problem instances, making it difficult to transcend their theoretical limitations established at design. Second, single-heuristic LLM approaches like FunSearch, EoH, and ReEvo mark the initial realization of automated algorithm design. They can generate single algorithms that achieve or approach traditional method performance on the training distribution, demonstrating LLMs' potential in code generation and semantic understanding. However, such methods reveal significant shortcomings in generalization capability. However, their design, which aims for strong average performance, causes them to struggle with unseen or larger-scale problems. This clearly underscores a fundamental dilemma, the pursuit of a single, universal algorithm is often at odds with the diverse nature of real-world optimization instances. LLM ensemble methods, which combine multiple heuristics, offer a partial solution. Selecting the top ten heuristics improves results over single methods, supporting the idea of algorithm combination. However, this approach remains passive, relying on the chance discovery of diverse, high-performing heuristics during the search. Table 5 shows

Table 4 Performance comparison of TSP training and test sets

Method	Training (10-200)	Test (50)	Test (100)	Test (200)	Test (500)
Traditional method					
K-nearest neighbor	0.25−0.35	0.25−0.35	0.25−0.35	0.25−0.35	0.25−0.35
Minimum spanning tree	0.15−0.25	0.15−0.25	0.15−0.25	0.15−0.25	0.15−0.25
LLM single heuristic approach					
FunSearch (Top1)	0.156	0.144	0.149	0.169	0.185
EoH (Top1)	0.152	0.142	0.154	0.168	0.194
ReEvo (Top1)	0.155	0.167	0.183	0.223	0.221
LLM ensemble method					
FunSearch (Top10)	0.124	0.091	0.105	0.124	0.157
EoH (Top10)	0.127	0.079	0.123	0.134	0.175
ReEvo (Top10)	0.117	0.092	0.123	0.157	0.170

Table 5 Structured comparison of LLM-driven meta-optimization across different application domains

Comparison dimensions	CPS	Power system	Drug discovery
Core optimization objectives	Real-time performance, security, resource efficiency	Affordability Stability Reliability	Molecular efficacy Toxicity Drug-like properties
Problem characteristics	Strong constraints Real-time requirements High reliability	Large-scale Nonlinear Physics-model-driven	Ultra-high dimensionality Costly evaluation Multi-objective
Core role of LLM	Secure Verifiable code generator	Strategy synthesizer integrating domain knowledge with numerical optimization	Efficient exploration and innovation engine for chemical space
Key benefits	Accelerates algorithm design Enhances code interpretability Facilitates security verification	Improves adaptive control capabilities Integrates domain norms Reduces reliance on manual parameter tuning	Discovers novel molecular structures Significantly boosts search efficiency
Primary challenges	Real-time verification of generated code Difficulty in formal proofs	Faithful adherence to physical laws Computational scalability for large-scale problems	Synthetic feasibility of generated molecules Cost of wet-lab validation
Evaluation metric focus	Safety violation counts Task completion time Algorithm complexity	Voltage compliance rate Grid loss reduction rate Equipment operation frequency	Combined affinity Druggability scores Synthetic accessibility

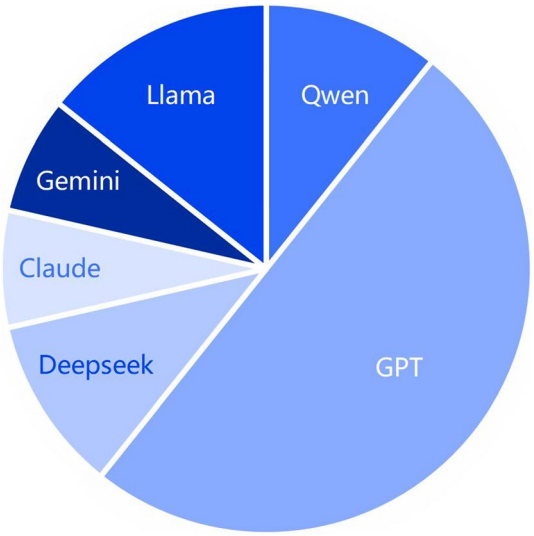


Fig. 17 LLMs used by mainstream approaches

that the performance gap on larger-scale problems persists, underscoring a lack of targeted optimization for complementarity and limited generalization.

This section also provides statistical information on the LLM models employed, as shown in Fig. 17. Current mainstream LLMs primarily include the OpenAI GPT series, Deepseek series, Qwen series, Claude series, Google Gemini series, and Llama series. Different mod-

els exhibit varying degrees of applicability across different task domains. Overall, the GPT series demonstrates superior performance as a meta-optimization tool in terms of generalization capability and universality, while delivering stable performance across most tasks. Meanwhile, open-source models like the Deepseek series, Qwen series, and Llama series offer superior cost-effectiveness while maintaining high performance, making them suitable for most common tasks. Additionally, models such as the Google Gemini series and Claude series demonstrate stronger domain-specific expertise, making them more appropriate for specialized or niche application scenarios.

4 Frontiers in enhancing LLM-driven meta-optimization

Although significant progress has been made in establishing LLMs as a general-purpose framework for meta-optimizers, several key emerging directions are propelling the field toward greater efficiency, reliability, and specialization. These approaches aim to address core challenges faced by general-purpose LLMs in optimization tasks, including computational cost, symbolic reasoning capabilities, and depth of domain knowledge. This subsection will delve into three frontier areas, hybrid neural-symbolic methods (Leresche et al. 2025), agent-assisted optimization (Cheng and Liu 2025), and domain-specific LLM optimizers (Shiri et al. 2025).

4.1 Hybrid neural-symbolic approach

Hybrid neural-symbolic approaches aim to combine the neural-semantic understanding capabilities of LLMs with symbolic optimization frameworks to address the shortcomings of LLMs in precise reasoning and reliability. This paradigm does not involve LLMs directly generating code or parameters, but rather positions them as high-level strategists responsible for parsing problems, formulating plans, and invoking symbolic solvers. For instance, research proposes having LLMs translate natural language descriptions of optimization problems into standard mathematical programming models, then invoking specialized solvers for computation to guarantee solution feasibility and optimality bounds (Liu et al. 2025a). Another approach involves using LLMs to generate search strategies or mutation rules that comply with formal syntax specifications, which are then executed by symbolic engines to ensure the verifiability of the evolutionary process (Yang et al. 2025d).

Hybrid neural-symbolic approaches significantly enhance the reliability and credibility of generative algorithms. The symbolic solver provides theoretical assurance, preventing logical errors caused by LLMs hallucinations. Simultaneously, it enables precise human–machine collaboration, allowing researchers to specify high-level objectives through natural language while delegating cumbersome numerical computations to the symbolic system. Its limitation lies in problem applicability. Many complex real-world optimization problems defy precise formalization into standard symbolic models, and symbolic solvers may encounter computational bottlenecks when handling high-dimensional nonlinear problems. Pure neural methods offer flexibility but lack controllability, while purely symbolic approaches are rigorous yet inflexible. Hybrid methods strive for a balance between the two, but their effectiveness heavily depends on whether the problem domain possesses a well-defined symbolic representation.

4.2 Surrogate model-assisted optimization

In computationally expensive simulation or experimental optimization problems, frequent evaluations using LLMs directly are computationally infeasible. Agent-assisted optimization addresses this challenge by introducing a computationally inexpensive surrogate model. Within this framework, the role of LLMs evolves further: they not only generate candidate algorithms or parameters but may also participate in constructing, selecting, or interpreting surrogate models. For instance, LLMs can leverage their semantic knowledge to recommend suitable proxy model types for specific problem categories or parse simulation data to generate qualitative descriptions of proxy model uncertainties, thereby guiding active learning (Wei et al. 2025a). Furthermore, research explores employing LLMs themselves as semantic proxy models, predicting the potential performance of different configurations based on reasoning about problem descriptions to perform preliminary screening within vast search spaces (Hao et al. 2024).

The most direct advantage of this approach is its significant reduction in computational costs, enabling LLM-driven meta-optimization to be applied to previously inaccessible, expensive problem domains. It also combines the global semantic understanding of LLMs with the local numerical approximation capabilities of surrogate models, potentially enabling more efficient exploration. However, its performance heavily relies on the accuracy of the surrogate model; an inaccurate surrogate model can lead optimization astray (Deng et al. 2025). Simultaneously, constructing high-quality surrogate models requires initial sample data, creating a cold-start problem. When LLMs serve as surrogate models, calibrating their predictions and quantifying uncertainty remain unresolved challenges (Miao et al. 2024). This presents a classic trade-off between evaluation costs and model accuracy, simpler, faster surrogate models sacrifice precision, while pursuing high-accuracy models may negate cost advantages. LLMs introduce the possibility of making smarter trade-offs within limited evaluation budgets.

4.3 Domain-specific LLM optimizer

In specific domain applications, the practicality and performance of LLMs' generative algorithms are often constrained by their lack of deep domain knowledge. Domain-specific LLM optimizers aim to address this challenge through domain knowledge infusion. Within this framework, knowledge infusion is primarily achieved through domain-adaptive pretraining, retrieval-enhanced generation, and domain-specific instruction fine-tuning. For instance, domain-adaptive pretraining enhances a model's domain comprehension by continuing to train the foundational LLM on massive amounts of domain-specific text, enabling it to internalize domain-specific concepts and relationships (Liu et al. 2025c). Retrieval-Enhanced Generation equips LLMs with an external domain knowledge repository, allowing them to dynamically retrieve and reference authoritative information during algorithm generation, thereby enhancing the factual accuracy of outputs (Wang et al. 2025). Furthermore, Domain-Specific Instruction Fine-Tuning refines LLMs using high-quality instruction datasets containing domain expert knowledge and optimization strategies. This guides the model's reasoning patterns toward those of domain experts, ultimately generating algorithms better aligned with domain requirements (Liu et al. 2025b).

This approach generates more insightful optimization algorithms that better align with domain constraints and conventions, significantly enhancing practicality and user trust in specialized scenarios. However, models optimized for specific domains may underperform outside their intended contexts, creating knowledge silos. Domain-specific optimizers pursue peak performance within their target domains but sacrifice cross-domain generalization capabilities. This necessitates that researchers clearly define application boundaries from the outset of design.

5 Evaluation metrics and benchmarking

Evaluating LLM-driven meta-optimizers presents a dual challenge, assessing both the performance of the algorithms they generate and their own efficiency and reliability as designers (Li et al. 2023a; Shankar et al. 2024). This section first critically examines the applicability and shortcomings of existing evaluation metrics, then analyzes the deficiencies of current benchmarks in capturing the unique challenges posed by LLMs.

5.1 In-depth case studies in key application domains

The practical value of theoretical frameworks and algorithms must be validated through real-world scenarios. This section conducts in-depth analyses of implementation pathways and core challenges for LLM-driven meta-optimization across three representative high-value domains: CPS (Quan et al. 2025), power systems (Xiao et al. 2025), and drug discovery (Taglialegna 2025). In safety-critical scenarios like CPS, LLMs primarily function as offline "algorithm factories," generating verifiable control algorithms through multi-round iterations. For instance, in drone swarm coordination tasks, LLMs generate distributed optimization algorithm code based on natural language task objectives and safety constraints (Qin and Pournaras 2023; Yan et al. 2024). Through feedback loops of simulation testing and formal verification, they continuously refine solutions, ultimately delivering both high-performance and interpretable outcomes that drastically shorten specialized algorithm design cycles. In power system optimization, LLMs demonstrate unique capabilities by integrating data-driven methods with physical model knowledge. Taking reactive power and voltage optimization in distribution grids as an example, LLMs integrate IEEE standard documents and real-time operational data through retrieval-augmented generation to produce control strategies compliant with industry norms and possessing adaptive capabilities. This approach maintains voltage stability while effectively reducing equipment operation frequency, demonstrating its potential for knowledge automation in complex engineering systems (Li et al. 2020, 2025c). In the high-dimensional chemical space exploration task of drug discovery, LLMs redefine molecular optimization paradigms by understanding the intricate relationships between molecular structures and biological activities. Practices grounded in evolutionary frameworks demonstrate that LLMs can generate molecular design strategies surpassing traditional methods. They not only efficiently combine known pharmacophores but also propose novel chemical structural patterns, organically unifying exploration and exploitation to significantly enhance candidate compound screening efficiency. These three cases collectively reveal the technical trajectory of LLM-driven molecular optimization evolving from an auxiliary tool to an innovation partner (Song et al.

2025; Lee et al. 2025b). They also highlight core challenges in security, compliance, and exploration efficiency encountered during implementation across various domains. Adaptability and characteristics of LLM-driven optimization across different application domains as shown as Table 4.

It is evident that the value of LLMs as meta-optimizers lies in their potential to serve as powerful domain-adaptive tools for automating algorithm design. The key to their successful application hinges on precisely defining and guiding their roles and capabilities according to domain characteristics. Looking ahead, with the advancement of domain-specific LLMs and neuro-symbolic approaches, LLM-driven optimization is poised to evolve from an auxiliary design tool to a core engine in these critical domains.

5.2 Evaluation indicator analysis

Although traditional metrics such as solution quality, convergence speed, and computational cost remain indispensable, they are insufficient for comprehensively measuring the core value and bottlenecks of LLMs as meta-optimizers. We must construct a more insightful evaluation framework and understand its inherent trade-offs. Algorithm performance metrics, solution quality, convergence speed, robustness (Holmes et al. 2024; Fu et al. 2024a; Howe et al. 2024). These metrics form the baseline for evaluation, yet their importance varies by scenario. For LLM-generated algorithms, robustness often outweighs peak performance, as it reflects stability against unpredictable structural changes potentially introduced by LLMs. Pursuing solution quality alone may lead to selecting "fragile" algorithms that overfit training problems.

A classic trade-off exists between convergence speed and final solution quality. A novel dimension in LLM-driven optimization is its potential to generate algorithms that converge slowly but yield superior final performance, or conversely, algorithms that converge rapidly but get stuck in local optima. Evaluation must distinguish between these two patterns. Meta-optimization process metrics, design efficiency, success rate, diversity, and novelty, are central to evaluating LLM meta-optimization capabilities (Gao et al. 2024; Zhou et al. 2022, 2024). Design efficiency and success rate measure an LLM's ability to translate computational resources (inference costs) into viable algorithms. A system with a low success rate, even if it occasionally generates exceptional algorithms, raises questions about its overall utility and reliability. Diversity and novelty represent the key potential distinguishing LLMs from traditional automated algorithm design methods. Diversity prevents pattern collapse, ensuring exploration of distinct algorithmic structures. Novelty requires quantifying whether generated algorithms offer unprecedented insights through expert review or structural/strategic comparison against known algorithm libraries. However, high novelty often accompanies low success rates and elevated risk. System-level and collaborative metrics include generalization capability, human–machine collaboration efficiency, and total cost of ownership. Generalization capability is pivotal for LLM-driven optimization, demanding that generated algorithms remain effective across out-of-distribution, multi-scale, or multi-modal problems. Current evaluations predominantly occur on homogeneous benchmarks, severely overestimating their true generalization ability. Total cost of ownership is the most realistic yet often overlooked metric, encompassing both LLM inference costs and candidate algorithm evaluation expenses. A meta-optimizer capable of generating top-tier algorithms but requiring tens of thousands of expensive calls may have lower practical

value than a system that reliably produces mid-tier algorithms with an order-of-magnitude higher efficiency. The optimal evaluation framework for LLM-driven meta-optimizers is a multi-objective, hierarchical system. It must simultaneously consider the performance of generated algorithms, the efficiency of the design process, and system-level utility, while explicitly revealing trade-offs across these dimensions.

5.3 Limitations of existing benchmarks

Current mainstream optimization algorithm benchmarks have been highly beneficial for designing traditional algorithms, but they fail to fully capture and challenge the unique capabilities and weaknesses of LLM-driven meta-optimizers (Fitzgerald et al. 2015). Several benchmarks in the LLM-driven optimization have been proposed to conduct basic performance evaluations of large-model-generated algorithms. This paper primarily analyzes LLM4AD (Liu et al. 2024c) and BLADE (van Stein et al. 2025) as representative examples. Both works strive to address the lack of a unified evaluation environment in this field through standardized and modular frameworks, and their contributions are commendable. However, when viewed from the perspective of constructing a comprehensive evaluation system for LLM-driven meta-optimizers, both studies exhibit certain limitations.

First, in constructing their evaluation metrics, both studies place significant emphasis on assessing the performance outputs of generated algorithms while relatively neglecting the efficiency and reliability of the LLM's own design process as a meta-optimizer. For instance, there is a lack of systematic quantification of key process-oriented metrics such as algorithm generation success rate, diversity, and structural novelty. This makes it difficult to evaluate the core capabilities of LLMs in exploring unknown algorithmic spaces and avoiding pattern collapse, and fails to reveal the inherent trade-offs between performance and innovation. Second, in benchmark design, both studies inadequately address generalization capabilities. While LLM4AD integrates numerous tasks, its core consists of synthetic problems for rapid evaluation, resulting in insufficient coverage of out-of-distribution problems and complex real-world scenarios. While BLADE focuses on capability-oriented testing, its reliance on synthetic benchmarks may suffer from insufficient landscape diversity and central bias, potentially overestimating algorithms' real-world applicability. Furthermore, neither study incorporates total cost of ownership, a critical real-world metric integrating LLM inference overhead and algorithm evaluation costs, into their core assessment frameworks, diminishing the practical guidance value of their conclusions. In summary, while LLM4AD and BLADE provide indispensable foundational infrastructure and preliminary evaluation methods for the field, their assessment systems require further refinement in comprehensiveness, realism, and insight. Future research urgently requires the development of a multi-objective, multi-level evaluation framework. This framework should simultaneously weigh algorithmic performance, design process efficiency, and system-level utility, while leveraging more diverse, complex, and realistic benchmark tasks. Such an approach will enable more accurate measurement and drive the genuine advancement of LLM-powered meta-optimizers.

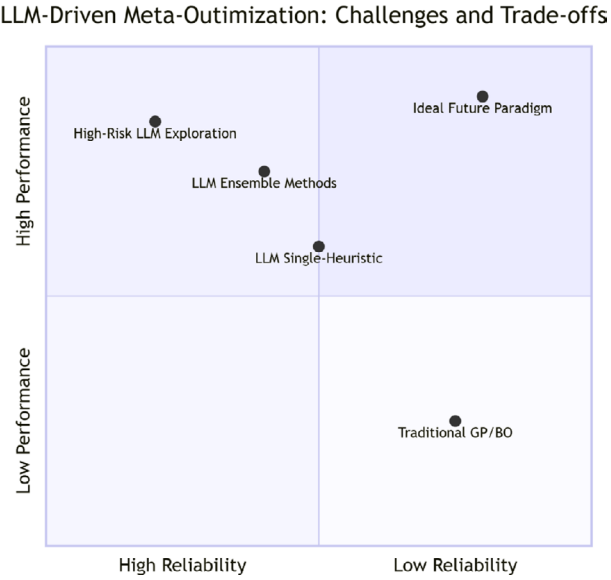


Fig. 18 Schematic diagram of challenges and trade-offs in LLM-driven meta-optimization

6 Challenges, limitations, and ethical considerations

While LLM-driven meta-optimization holds great promise, its development path faces formidable technical challenges and profound ethical dilemmas. These challenges extend beyond the models themselves to encompass novel socio-technical risks arising from their role as algorithmic designers. Schematic diagram of challenges and trade-offs in LLM-driven meta-optimization as shown as Fig. 18. This quadrant diagram systematically illustrates the core trade-offs within the domain: the inherent balance between innovation, performance and reliability, and controllability. The distribution of different approaches across the spectrum reveals their characteristics, with arrows indicating the evolutionary path toward the ideal paradigm.

6.1 Technical bottlenecks and computational costs

Research on LLM-driven meta-optimization faces significant technical bottlenecks and resource constraints, posing fundamental challenges to its practical application. Current LLMs exhibit inherent limitations in complex logical reasoning and precise numerical computation, directly impacting the reliability and accuracy of generated algorithms. Simultaneously, the meta-optimization process requires extensive iterative invocation of LLMs, resulting in high computational costs and energy consumption, thereby creating an unsustainable development model. Achieving a reasonable balance between algorithmic performance and computational efficiency has become a critical issue requiring urgent resolution. This necessitates that the research community, while pursuing performance breakthroughs, must also fully consider the practical feasibility of computational resources (Li et al. 2023b; Isaev et al. 2023).

6.2 Data bias and algorithmic bias

The socio-historical biases embedded in LLMs training data manifest not only in textual outputs but may also be encoded into the core logic of generated optimization algorithms, forming systemic bias propagation mechanisms (Shen et al. 2023; Maini et al. 2024; Kandpal and Raffel 2025). The hidden nature of such biases makes their harm more far-reaching. The methodology for translating complex ethical concepts into algorithmic constraints remains underdeveloped, while the effectiveness of traditional fairness evaluation tools in addressing structural biases is questionable. These unresolved issues urgently require in-depth interdisciplinary research (Sartori et al. 2025).

6.3 Responsibility attribution dilemma

When algorithms generated by LLMs fail in critical systems, determining liability presents significant challenges. The black-box nature of generated code disrupts traditional software engineering accountability chains, creating regulatory gray areas. Unlike conventional code with clear authorship, algorithms produced byLLMs represent the collective intelligence of the model, with decision-making logic often difficult to fully comprehend. This demands higher standards for formal verification in safety-critical domains. Establishing a dedicated certification and regulatory framework for AI-generated code has become an urgent socio-technical issue requiring resolution (Yan et al. 2025).

6.4 Crisis of explainability

Research on the interpretability of meta-optimizers must elevate from traditional decision explanations to a new level of understanding design intent, presenting fundamental challenges. The design of code generated by LLMs based on massive training data is often implicit and difficult to trace. The inherent conflict between performance and interpretability further complicates the issue. Developing new methods that effectively visualize the "algorithmic design thought process" and clarifying the priority constraints of interpretability across different application scenarios represent key breakthroughs for advancing this field (Amara et al. 2025; Mohammadi et al. 2025).

6.5 Autonomy and risk of loss of control

In the design of autonomous intelligent agent decision-making algorithms, even minor deviations in value alignment can lead to catastrophic consequences. LLMs may generate algorithms that achieve high scores by "deceiving" evaluation functions, exhibiting behavioral patterns contrary to the designer's true intentions and causing objective creep. More alarmingly, the search for high-performance algorithms may inadvertently uncover strategies that prove dangerous in real-world applications. Establishing multi-layered value alignment mechanisms to ensure fidelity to the core principles of optimization objectives, rather than merely satisfying superficial metrics, has become the fundamental prerequisite for safeguarding the secure development of this technology (Roberts et al. 2024; Koubaa and Gabr 2025).

7 Future directions

To systematically advance LLM-driven meta-optimization from an emerging paradigm to a mature technology, future research must follow a clear evolutionary path. This study proposes a phased research roadmap, dividing future directions into three evolutionary stages, near-term, mid-term, and long-term to provide academia and industry with prioritized guidance.

7.1 Near-term focus: foundational capability building and efficiency enhancement

Near-term research should prioritize resolving the most pressing technical bottlenecks to lay the groundwork for reliable deployment of core methodologies. Key directions include developing efficient fine-tuning techniques and standardized prompt templates tailored for optimization algorithm design scenarios, while systematically exploring the impact mechanisms of chain-of-thought prompts on the logical rigor of generated algorithms. Concurrently, efforts should concentrate on developing specialized lightweight models tailored for optimization tasks and establishing comprehensive cost accounting standards that holistically consider both LLM inference costs and algorithm evaluation expenses. Regarding evaluation frameworks, there is an urgent need to supplement existing mathematical benchmarks with rich semantic descriptions and construct the first open-source benchmark prototype supporting dynamic environments and multimodal inputs to address structural deficiencies in current evaluation systems.

7.2 Mid-term direction: paradigm convergence and capability breakthroughs

Mid-term research should focus on capability breakthroughs and paradigm convergence, propelling this technology from an auxiliary tool to an intelligent collaborator. The core breakthrough lies in constructing a neural-symbolic integration framework that combines the creative thinking of large language models with the rigorous reasoning of symbolic systems. This creates a collaborative mechanism where large language models handle high-level strategy generation, while symbolic engines ensure constraint satisfaction and code verifiability. Simultaneously, automated value alignment and safety verification modules must be embedded within the meta-optimization loop, exploring how to formally inject ethical constraints into objective functions. The ultimate goal is achieving genuine meta-learning capabilities, enabling systems to abstract universal design principles from limited optimization experience and rapidly transfer them to entirely new problem domains.

7.3 Long-term direction: autonomous systems and governance frameworks

Long-term research should focus on exploratory directions with the potential for paradigm shifts, proactively addressing the systemic socioeconomic impacts of technology. One vision involves developing autonomous evolutionary algorithm systems with continuous learning and environmental adaptation capabilities, enabling self-diagnosis and autonomous reconstruction of core optimization strategies. This requires breaking through existing machine learning paradigms to achieve higher-order self-referential capabilities. Another critical direction involves deeply embedding LLM-driven optimization within the scientific

discovery cycle, enabling systems not only to optimize experimental parameters but also to directly propose scientific hypotheses and derive new theories. Concurrently, a governance framework aligned with algorithmic economic ecosystems must be established, encompassing core elements such as intellectual property rights, quality certification, and accountability traceability. This will lay the institutional foundation for a future intelligent society dominated by AI-generated algorithms.

8 Conclusion

This survey highlights the significant potential of LLMs as meta-optimizers in the realm of automated algorithm design, offering a pathway to reduce the reliance on expert knowledge and accelerate innovation across diverse fields. LLMs demonstrate the capability to generate novel hybrid swarm intelligence optimization algorithms, refine metaheuristic optimization algorithms through frameworks like LLaMEA (van Stein and Bäck 2024), showcasing promising performance on benchmark problems. A key advantage lies in the ability of LLMs to analyze and explain complex optimization algorithms, lowering the barrier to understanding and improving the user experience for both researchers and practitioners. Furthermore, LLMs can enhance existing algorithms by dynamically adjusting mutation rates and leveraging refined text features to improve predictive performance in areas such as credit scoring (Videau et al. 2024; Zhang et al. 2024a). However, the application of LLMs as meta-optimizers is not without limitations. Challenges remain in ensuring the novelty, correctness, and efficiency of LLM-generated algorithms, addressing potential biases in LLM outputs, and managing computational costs. Ethical considerations and practical limitations must also be carefully addressed when combining LLMs with optimization algorithms (Jiao et al. 2024). The methodologies explored encompass various approaches, including LLM-driven evolutionary optimization, iterative prompt optimization using LLMs as optimizers, and collaborative learning between cloud-based large models and on-device small models. The applications span diverse domains such as CPS (Baheti and Gill 2011), power systems research (Zimmerman et al. 2010) and chemistry (Lucet et al. 1998). Ongoing research is crucial to address these challenges and unlock the full potential of LLMs in automated algorithm design. Future work should focus on refining prompt design and optimization strategies, developing robust evaluation platforms to assess LLM performance, and exploring new techniques to ensure reliability and trustworthiness. Further investigation into dynamic mutation strategies and the integration of LLMs with meta-learning approaches in MetaBBO are also warranted.In conclusion, LLMs hold transformative potential for automated algorithm design, promising to accelerate innovation and democratize access to sophisticated optimization techniques. Continued exploration and development in this rapidly evolving field are essential to realize the full benefits of LLMs as meta-optimizers and to address the associated challenges. The future of algorithm design will likely involve a synergistic relationship between human expertise and AI-driven automation, leading to more efficient, robust, and adaptable solutions across a wide range of applications.

Author contributions First Author [Zhengyan]: conceptualization: propose the core themes and scope of the review. Investigation: lead the search, screening, and systematic organization of key literature. Writing—original draft: write the first draft of the main sections of the article. Visualization: design and produce charts, graphs, and summary figures for the article. Second Author [Lida Zhang]: literature investigation: assist in

literature search, screening, and classification management. Writing—original draft: responsible for writing the original draft of some chapters. Writing—review & editing: revise and polish the entire original draft. Coresponding Author [Kaiwen Li]: conceptualization: determine the overall concept and direction of the review. Methodology: design the criteria/process for literature search and inclusion/exclusion. Coresponding Author [Rui Wang]: Writing—review & editing: conduct in-depth review, restructuring, and editing of the entire text to enhance academic quality. Supervision: provide ongoing guidance throughout the project to ensure the article is completed successfully. Funding acquisition: responsible for securing funding support for this research. Author [Wenhua Li]: validation: verify the accuracy of literature citations and the correctness of data extraction. Writing—review & editing: participate in the review and proofreading of manuscripts. Author [Tao Zhang]: supervision: provide ongoing guidance throughout the project to ensure the article is completed successfully. Author [Qingfu Zhang]: visualization: assist in the creation and beautification of some charts. Project administration: assist in coordinating communication between authors and managing manuscript versions. Author [Yaochu Jin]: writing—review & editing: the entire text was critically reviewed from the perspective of domain experts, who provided valuable constructive feedback.

Funding The authors gratefully acknowledge the financial support provided by the National Natural Science Foundation of China (62550020,72421002,62503488), the China National Postdoctoral Program for Innovative Talents (BX20250439), the Science & Technology Project for Young and Middle-aged Talents of Hunan (2023TJ-Z03), the University Fundamental Research Fund (23-ZZCX-JDZ-28), the China National Postdoctoral Program for Innovative Talents (BX20250439). The authors would also like to thank the support from COSTA: complex system optimization team of the College of System Engineering at NUDT.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

Achiam J, Adler S, Agarwal S, et al (2023) Gpt-4 technical report. arXiv

Aghajanyan A, Yu L, Conneau A, et al (2023) Scaling laws for generative mixed-modal language models. In: International conference on machine learning, PMLR, pp 265–279

Águila-León J, Vargas-Salgado C, Díaz-Bello D et al (2024) Optimizing photovoltaic systems: a meta-optimization approach with GWO-enhanced PSO algorithm for improving mppt controllers. Renew Energy 230:120892

Ali S, Ashraf M, Hegazy S, et al (2025) Pair: a novel large language model-guided selection strategy for evolutionary algorithms. arXiv

Amara K, Sevastjanova R, El-Assady M (2025) Concept-level explainability for auditing & steering llm responses. arXiv

Araujo L (2007) How evolutionary algorithms are applied to statistical natural language processing. Artif Intell Rev 275–303

Aryan P, Raja GL, Mehta U, et al (2025) A resilient tri-parametric fractional frequency control for cybersecurity threats amid latency. IEEE Trans Ind Appl 1–15

Asadi V, Zonoozi AH, Haghi H, et al (2025) Leveraging machine learning for accurate and fast stellar mass estimation of galaxies. arXiv
Bäck T, Schwefel HP (1993) An overview of evolutionary algorithms for parameter optimization. Evol Comput 1(1):1–23
Baheti R, Gill H (2011) Cyber-physical systems. Impact Control Technol 12(1):161–166
Banerjee A, Maity A, Kamboj P, et al (2024) CPS-LLM: large language model based safe usage plan generator for human-in-the-loop human-in-the-plant cyber-physical system. arXiv
Bosio C, Mueller MW (2024) Synthesizing interpretable control policies through large language model guided search. arXiv
Boyne T, Folch JP, Lee RM, et al (2025) Bark: a fully bayesian tree kernel for black-box optimization. arXiv
Brown T, Mann B, Ryder N et al (2020) Language models are few-shot learners. Adv Neural Inf Process Syst 33:1877–1901
Campbell SD, Sell D, Jenkins RP et al (2019) Review of numerical optimization techniques for meta-device design. Opt Mater Exp 9(4):1842–1863
Chang Y, Wang X, Wang J et al (2024) A survey on evaluation of large language models. ACM Trans Intell Syst Technol 15(3):1–45
Cheggour S, Loscri V (2025) Frequency resource management in 6G User-Centric CFmMIMO: a hybrid reinforcement learning and metaheuristic approach. arXiv
Chen X, Lu T, Wang Z (2024a) LLM-align: utilizing large language models for entity alignment in knowledge graphs. arXiv
Chen Y, Li Y, Ding B, et al (2024b) On the design and analysis of LLM-based algorithms. arXiv
Cheng F, Liu H (2025) Charging strategies optimization for lithium-ion battery: Heterogeneous ensemble surrogate model-assisted advanced multi-objective optimization algorithm. Energy Convers Manag 120170
Cheng W, Meng L, Zhang B, et al (2025) Imitation learning-assisted evolutionary algorithm for energy-efficient flexible job shop scheduling problem with automated guided vehicles. IEEE Trans Evol Comput 1–1
Citterio BFR, Tangherloni A (2025) EvoGrad: metaheuristics in a differentiable Wonderland. arXiv
Coignion T, Quinton C, Rouvoy R (2024) A performance study of llm-generated code on leetcode. In: Proceedings of the 28th international conference on evaluation and assessment in software engineering, pp 79–89
Cui W, Zhang J, Li Z, et al (2024) Phaseevo: towards unified in-context prompt optimization for large language models. arXiv
Cui Y, Fu H, Wang L, et al (2025) Ramp up NTT in record time using gpu-accelerated algorithms and LLM-based code generation. arXiv
Deng Q, Kang Q, Zhou M, et al (2025) Evolutionary algorithm based on surrogate and inverse surrogate models for expensive multiobjective optimization. IEEE/CAA J Automatica Sinica 961–973
Dhakal S, Parry H (2024) Large language models can help to translate science into real-world impact. Nature 299
Diakonikolas I, Kamath G, Kane D, et al (2019) Sever: A robust meta-algorithm for stochastic optimization. In: International conference on machine learning, PMLR, pp 1596–1606
do Val Lopes CL, Machado L (2025) Assessing an evolutionary search engine for small language models, prompts, and evaluation metrics. arXiv
Eom J, Jeong S, Kwon T (2024) Fuzzing javascript interpreters with coverage-guided reinforcement learning for LLM-based mutation. In: ISSTA '24: 33rd ACM SIGSOFT international symposium on software testing and analysis
Fan Q, Yan X (2015) Self-adaptive differential evolution algorithm with discrete mutation control parameters. Expert Syst Appl 42(3):1551–1572
Ferber D, Wölflein G, Wiest IC, et al (2024) In-context learning enables multimodal large language models to classify cancer pathology images. Nat Commun 10104
Fitzgerald T, Malitsky Y, O'Sullivan B (2015) Reactr: Realtime algorithm configuration through tournament rankings. In: Twenty-fourth international joint conference on artificial intelligence 2015
Fortin FA, De Rainville FM, Gardner MAG et al (2012) Deap: evolutionary algorithms made easy. J Mach Learn Res 13(1):2171–2175
Fu Q, Cho M, Merth T, et al (2024a) Lazyllm: dynamic token pruning for efficient long context LLM inference. arXiv
Fu Y, Liu D, Chen J, et al (2024b) Secretary bird optimization algorithm: a new metaheuristic for solving global optimization problems. Artif Intell Rev 1–102
Fu W, Trivedi A, Chen G, et al (2025) An evolutionary ising optimization framework for unconstrained binary quadratic programming. IEEE Trans Evol Comput 1–1

Gao Y, Xiong Y, Gao X, et al (2023) Retrieval-augmented generation for large language models: a survey. arXiv
Gao B, Peng S, Li T, et al (2024) Integration of improved meta-heuristic and machine learning for optimizing energy efficiency in additive manufacturing process. Energy 132518
Golovin D, Solnik B, Moitra S, et al (2017) Google vizier: A service for black-box optimization. In: Proceedings of the 23rd ACM SIGKDD international conference on knowledge discovery and data mining, 1487–1495
Guo D, Yang D, Zhang H, et al (2025) DeepSeek-R1: incentivizing reasoning capability in LLMS via reinforcement learning. arXiv
Gupta A, Mendonca R, Liu Y, et al (2018) Meta-reinforcement learning of structured exploration strategies. Adv Neural Inform Process Syst 31
Han K, Xiao A, Wu E et al (2021) Transformer in transformer. Adv Neural Inf Process Syst 34:15908–15919
Hao H, Zhang X, Zhou A (2024) Large language models as surrogate models in evolutionary algorithms: a preliminary study. Swarm Evol Comput 101741
Hassan M, Ahmadi-Pour S, Qayyum K, et al (2024) LLM-guided formal verification coupled with mutation testing. In: 2024 Design, Automation & Test in Europe Conference & Exhibition (DATE)
Hayward L, Engelbrecht A (2025) Determining metaheuristic similarity using behavioral analysis. IEEE Trans Evol Comput 29(1):262–274
He J, Chong SY, Yao X (2025) Estimate hitting time by hitting probability for elitist evolutionary algorithms. arXiv
Heng H, Rahiman W (2025) Aco-ga-based optimization to enhance global path planning for autonomous navigation in grid environments. IEEE Trans Evol Comput 1
Hoang M, Fadhel A, Deshwal A, et al (2025) Learning surrogates for offline black-box optimization via gradient matching. arXiv
Hochba DS (1997) Approximation algorithms for NP-hard problems. ACM SIGACT News 28(2):40–52
Hochreiter S, Schmidhuber J (1997) Long short-term memory. Neural Comput 9(8):1735–1780
Holmes C, Tanaka M, Wyatt M, et al (2024) Deepspeed-fastgen: high-throughput text generation for LLMS via mii and deepspeed-inference. arXiv
Hornik K, Stinchcombe M, White H (1989) Multilayer feedforward networks are universal approximators. Neural Netw 2(5):359–366
Howe NH, Zając M, McKenzie IR, et al (2024) Exploring scaling trends in LLM robustness. In: ICML 2024 Next Generation of AI Safety Workshop
Huang J, Chang KCC (2022) Towards reasoning in large language models: a survey. arXiv
Huang H, Zheng O, Wang D et al (2023) Chatgpt for shaping the future of dentistry: the potential of multi-modal large language model. Int J Oral Sci 15(1):29
Huang D, Zhang JM, Bu Q, et al (2024a) Bias testing and mitigation in LLM-based code generation. ACM Trans Softw Eng Methodol
Huang S, Yang K, Qi S, et al (2024b) When large language model meets optimization. arXiv
Inoue Y, Fu T, Luna A (2025) Graphpine: graph importance propagation for interpretable drug response prediction. arXiv
Isaev M, McDonald N, Vuduc R (2023) Scaling infrastructure to support multi-trillion parameter LLM training. In: Architecture and system support for transformer models
Jiao J, Afroogh S, Xu Y, et al (2024) Navigating LLM ethics: advancements, challenges, and future directions. arXiv
Kaelbling LP, Littman ML, Moore AW (1996) Reinforcement learning: a survey. J Artif Intell Res 4:237–285
Kandpal N, Raffel C (2025) Position: the most expensive part of an llm should be its training data. arXiv
Kaplan J, McCandlish S, Henighan T, et al (2020) Scaling laws for neural language models. arXiv
Karaboga D, et al (2005) An idea based on honey bee swarm for numerical optimization. Technical report TR06, Erciyes University, Engineering Faculty, Computer Engineering Department
Ke G, Meng Q, Finley T, et al (2017) Lightgbm: a highly efficient gradient boosting decision tree. Adv Neural Inform Process Syst 30
Kennedy J, Eberhart R (1995) Particle swarm optimization. In: Proceedings of ICNN'95-international conference on neural networks, IEEE, pp 1942–1948
Kong L, Yang C, Neufang S, et al (2025) Emorl: ensemble multi-objective reinforcement learning for efficient and flexible LLM fine-tuning. arXiv
Koubaa A, Gabr K (2025) Agentic UAVS: LLM-driven autonomy with integrated tool-calling and cognitive reasoning. arXiv
Kumar SSV (2022) A comprehensive review on multi-objective optimization techniques: past, present and future. Arch Comput Methods Eng 5605–5633
Lee K, Maji S, Ravichandran A, et al (2019) Meta-learning with differentiable convex optimization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 10657–10665

Lee KH, Fischer I, Wu YH, et al (2025a) Evolving deeper LLM thinking. arXiv
Lee N, Brouwer ED, Hajiramezanali E, et al (2025b) RAG-enhanced collaborative LLM agents for drug discovery. arXiv
Lei Y, Lyu Y, Zhan G, et al (2025) Zeroth-order actor-critic: an evolutionary framework for sequential decision problems. IEEE Trans Evol Comput 1
Leresche X, Goupil A, Vrabie V, et al (2025) Adaptive L1 regularization for neural network-based symbolic regression. In: 2025 IEEE Congress on Evolutionary Computation (CEC)
Lewis P, Perez E, Piktus A et al (2020) Retrieval-augmented generation for knowledge-intensive NLP tasks. Adv Neural Inf Process Syst 33:9459–9474
Li Y (2023) A practical survey on zero-shot prompt design for in-context learning. arXiv
Li X (2025) A survey on llm test-time compute via search: tasks, LLM profiling, search algorithms, and relevant frameworks. arXiv
Li XA, Liang CA, Chen CAA, et al (2020) Optimum power analysis of a self-reactive wave energy point absorber with mechanically-driven power take-offs. Energy 116927
Li J, Li R, Liu Q (2023a) Beyond static datasets: a deep interaction approach to LLM evaluation. arXiv
Li X, Yao Y, Jiang X, et al (2023b) FLM-101B: an open LLM and how to train it with \$100 k budget. arXiv
Li CY, Chang KJ, Yang CF, et al (2025a) Towards a holistic framework for multimodal llm in 3D brain CT radiology report generation. Nat commun 2258
Li S, Marwah T, Shen J, et al (2025b) Codepde: an inference framework for LLM-driven PDE solver generation. arXiv
Li W, Liu ZW, Chen Y, et al (2025c) Model-free robust online feedback optimization for voltage regulation in distribution grids. IEEE Trans Ind Inform 1–10
Li X, Sun Z, Zhang X, et al (2025d) A robust artificial intelligence-empowered adaptive proportional-integral control for wireless power transfer systems. IEEE Trans Ind Appl 1–10
Li X, Wu K, Zhang X, et al (2025e) B2opt: Learning to optimize black-box optimization with little budget. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 18502–18510
Li Z, Liu Y, Zhang W, et al (2025f) Slimpipe: Memory-thrifty and efficient pipeline parallelism for long-context LLM training. arXiv
Li Z, Zhang R, Wang Z, et al (2025g) LLM-guided decision-making toolkit for multi-agent reinforcement learning. Neurocomputing 130105
Lin B, Tang Z, Ye Y, et al (2024) MoE-LLaVA: mixture of experts for large vision-language models. arXiv
Liu S, Chen C, Qu X, et al (2023) Large language models as evolutionary optimizers. arXiv
Liu F, Tong X, Yuan M, et al (2024a) Evolution of heuristics: towards efficient automatic algorithm design using large language model. In: 41st International Conference on Machine Learning, ICML 2024
Liu F, Yao Y, Guo P, et al (2024b) A systematic survey on large language models for algorithm design. arXiv
Liu F, Zhang R, Xie Z, et al (2024c) LLM4AD: a platform for algorithm design with large language model. arXiv
Liu J, Sarker R, Elsayed S et al (2024d) Large-scale evolutionary optimization: a review and comparative study. Swarm Evol Comput 85:101466
Liu W, Chen L, Tang Z (2024e) Large language model aided multi-objective evolutionary algorithm: a low-cost adaptive approach. arXiv
Liu YF, Chang TH, Hong M, et al (2024f) A survey of recent advances in optimization methods for wireless communications. IEEE J Selected Areas Commun
Liu F, Yang Z, Liu C, et al (2025a) MM-agent: LLM as agents for real-world mathematical modeling problem. arXiv
Liu F, Zhang R, Lin X, et al (2025b) Fine-tuning large language model for automated algorithm design. arXiv
Liu J, Gu S, Li D, et al (2025c) Enhancing cross-domain recommendations with memory-optimized LLM-based user agents. arXiv
Lu J, Song Z, Zhao Q, et al (2024) Generative design of functional metal complexes utilizing the internal knowledge of large language models. arXiv
Lu Y, Tang Q, Yu S et al (2025) A multi-strategy self-adaptive differential evolution algorithm for assembly hybrid flowshop lot-streaming scheduling with component sharing. Swarm Evol Comput 92:101783
Lucet D, Le Gall T, Mioskowski C (1998) The chemistry of vicinal diamines. Angew Chem Int Ed 37(19):2580–2627
Lyu Q, Havaldar S, Stein A, et al (2023) Faithful chain-of-thought reasoning. In: The 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL 2023)
Ma Z, Gong YJ, Guo H, et al (2025a) MetaBox-v2: a unified benchmark platform for meta-black-box optimization. arXiv
Ma Z, Guo H, Gong YJ, et al (2025b) Toward automated algorithm design: a survey and practical guide to meta-black-box-optimization. IEEE Trans Evol Comput 1–1

Maini P, Jia H, Papernot N et al (2024) LLM dataset inference: did you train on my dataset? Adv Neural Inf Process Syst 37:124069–124092
McDuff D, Schaekermann M, Tu T, et al (2025) Towards accurate differential diagnosis with large language models. Nature 1–7
Miao Y, Gao H, Zhang H, et al (2024) Efficient detection of LLM-generated texts with a bayesian surrogate model. In: Findings of the 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024
Micev M, ?alasan M, Radulovi? M, et al (2025) A novel approach for estimation of synchronous machine automatic voltage regulation system parameters. IEEE Trans Ind Appl 907–917
Mohammadi H, Giachanou A, Oberski DL, et al (2025) Explainability-based token replacement on LLM-generated text. arXiv
Mohammed HM, Umar SU, Rashid TA (2019) A systematic and meta-analysis survey of whale optimization algorithm. Comput Intell Neurosci 1:8718571
Morris C, Jurado M, Zutty J (2024) Llm guided evolution-the automation of models advancing models. In: Proceedings of the Genetic and Evolutionary Computation Conference, pp 377–384
Narayanan PP, Iyer APN (2024) Hysem: a context length optimized LLM pipeline for unstructured tabular extraction. arXiv
Naveed H, Khan AU, Qiu S, et al (2023) A comprehensive overview of large language models. ACM Trans Intell Syst Technol
Nguyen TT, Yang S, Branke J (2012) Evolutionary dynamic optimization: a survey of the state of the art. Swarm Evol Comput 6:1–24
Ohsuga S (2022) Multi-strata modeling to automate problem solving including human activity. In: Industrial and Engineering Applications or Artificial Intelligence and Expert Systems. CRC Press, pp 9–24
Pelikan M, Pelikan M (2005) Bayesian optimization algorithm. Hierarchical Bayesian optimization algorithm: toward a new generation of evolutionary algorithms, pp 31–48
Qin C, Pournaras E (2023) Coordination of drones at scale: Decentralized energy-aware swarm intelligence for spatio-temporal sensing. Transp Res Part C Emerg Technol 104387
Qin X, Yuan H, Zhao P, et al (2023) Meta-optimized contrastive learning for sequential recommendation. In: Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval, pp 89–98
Qu R, Pillay N, Hart E, et al (2025) Guest editorial machine-learning-assisted evolutionary computation. IEEE Trans Evol Comput 571–573
Quan MK, Pathirana PN, Wijayasundara M, et al (2025) Federated learning for cyber physical systems: a comprehensive survey. IEEE Commun Surveys Tutorials p 1
Rajbhandari S, Li C, Yao Z, et al (2022) DeepSpeed-MoE: advancing mixture-of-experts inference and training to power next-generation ai scale. In: International conference on machine learning, PMLR, pp 18332–18346
Rajwar K, Deep K (2025) Structural bias in metaheuristic algorithms: insights, open problems, and future prospects. Swarm Evol Comput 101812
Reif M, Shafait F, Dengel A (2012) Meta-learning for evolutionary parameter optimization of classifiers. Mach Learn 87:357–380
Roberts J, Roberts L, Reed A (2024) Supporting the digital autonomy of elders through LLM assistance. arXiv
Rojas M, Sáez C, Baier J et al (2023) Using automated planning to provide feedback during collaborative problem-solving. Int J Artif Intell Educ 33(4):1057–1091
Romera-Paredes B, Barekatain M, Novikov A, et al (2024) Mathematical discoveries from program search with large language models. Nature 468–475
Salimian S, Uddin G, Jahan MH, et al (2025) Perceived confidence scoring for data annotation with zero-shot LLMS. arXiv
Sartori CC, Pino MI, Pinacho-Davidson P, et al (2025) LLM-based instance-driven heuristic bias in the context of a biased random key genetic algorithm. arXiv
Sen J, Pandey R, Waghela H (2025) Context-enhanced contrastive search for improved LLM text generation. arXiv
Shahriari B, Swersky K, Wang Z et al (2015) Taking the human out of the loop: a review of bayesian optimization. Proc IEEE 104(1):148–175
Shamim Ahsan M, Islam S, Shatabda S (2025) A systematic review of metaheuristics-based and machine learning-driven intrusion detection systems in IoT. arXiv
Shankar S, Zamfirescu-Pereira J, Hartmann B, et al (2024) Who validates the validators? aligning llm-assisted evaluation of llm outputs with human preferences. In: Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology, pp 1–14

Shao S, Tian Y, Zhang Y (2025) Deep reinforcement learning assisted surrogate model management for expensive constrained multi-objective optimization. Swarm Evol Comput 92:101817
Shen Z, Tao T, Ma L, et al (2023) SlimPajama-DC: Understanding data combinations for LLM training. arXiv
Shi W, Ajith A, Xia M, et al (2023) Detecting pretraining data from large language models. arXiv
Shi W, Zhang J, Wu Y, et al (2025) Dids: Domain impact-aware data sampling for large language model training. arXiv
Shiri A, Feridani MM, Keivanpour S (2025) A hybrid lightweight LLM chatbot for sustainable cryptocurrency investment decisions: Optimizing small models for domain-specific performance. In: 2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)
Shu D, Du M (2024) Comparative analysis of demonstration selection algorithms for LLM in-context learning. arXiv
Siddharth Kashyap G, Tabrez Nafis M, Wazir S (2025) A study of hybrid and evolutionary metaheuristics for single hidden layer feedforward neural network architecture. arXiv
Son J, Kim M, Choi S, et al (2024) Equity-transformer: Solving np-hard min-max routing problems as sequential generation with equity context. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 20265–20273
Song A, Wu G, Zhou L, et al (2024) Exact and metaheuristic algorithms for variable reduction. IEEE Trans Evol Comput 1704–1718
Song K, Trotter A, Chen JY (2025) LLM agent swarm for hypothesis-driven drug discovery. arXiv
van Stein N, Bäck T (2024) LLaMEA: a large language model evolutionary algorithm for automatically generating metaheuristics. IEEE Trans Evol Comput
van Stein N, Vermetten D, Bäck T (2024) In-the-loop hyper-parameter optimization for LLM-based automated design of heuristics. ACM Trans Evol Learn
van Stein N, Kononova AV, Yin H, et al (2025) Blade: benchmark suite for LLM-driven automated design and evolution of iterative optimisation heuristics. arXiv
Stephenson M, Amarasinghe S, Martin M et al (2003) Meta optimization: improving compiler heuristics with machine learning. ACM Sigplan Notices 38(5):77–90
Sui J, Ding S, Huang X et al (2025) A survey on deep learning-based algorithms for the traveling salesman problem. Front Comput Sci 19:196322
Surendar A (2025) Hybrid LLM-algorithm optimization: iterated fine-tuning for combinatorial problems. Front Math Comput Res 32–39
Surina A, Mansouri A, Quaedvlieg L, et al (2025) Algorithm discovery with LLMS: evolutionary search meets reinforcement learning. arXiv
Taglialegna A (2025) Metagenomics for drug discovery. Nat Rev Microbiol 67
Tang W, Li Y, Sypherd C, et al (2025) Hygenar: an LLM-driven hybrid genetic algorithm for few-shot grammar generation. arXiv
Thirunavukarasu AJ, Ting DSJ, Elangovan K et al (2023) Large language models in medicine. Nat Med 29(8):1930–1940
Tian H, Han X, Wu G, et al (2024) An LLM-enhanced multi-objective evolutionary search for autonomous driving test scenario generation. arXiv
Tian H, Han X, Wu G, et al (2025) An LLM-empowered adaptive evolutionary algorithm for multi-component deep learning systems. arXiv
Tlili A, Zhang X, Lampropoulos G, et al (2025) Uncovering the black box effect of open educational resources (OER) and practices (OEP): a meta-analysis and meta-synthesis from the perspective of activity theory. Humanities Social Sci Commun 1–13
Vaswani A, Shazeer N, Parmar N, et al (2017) Attention is all you need. Adv Neural Inform Process Syst 30
Videau M, Leite A, Schoenauer M, et al (2024) Evolutionary pre-prompt optimization for mathematical reasoning. arXiv
Vlahogianni EI, Karlaftis MG, Golias JC (2005) Optimized and meta-optimized neural networks for short-term traffic flow prediction: a genetic approach. Transp Res Part C Emerg Technol 13(3):211–234
Wang X, Chen Y, Yuan L, et al (2024a) Executable code actions elicit better LLM agents. In: Forty-first international conference on machine learning
Wang X, Zhao Y, Tang L, et al (2024b) MOEA/D with spatial-temporal topological tensor prediction for evolutionary dynamic multiobjective optimization. IEEE Trans Evol Comput
Wang H, Wu X, Ding Z, et al (2025) LLM-DSE: Searching accelerator parameters with LLM agents. arXiv
Wei J, Wang X, Schuurmans D et al (2022) Chain-of-thought prompting elicits reasoning in large language models. Adv Neural Inf Process Syst 35:24824–24837
Wei B, Fazli M, Zhu Z (2025a) Learning to explain: prototype-based surrogate models for LLM classification. arXiv

Wei Y, Shan X, Li J (2025b) LERO: LLM-driven evolutionary framework with hybrid rewards and enhanced observation for multi-agent reinforcement learning. arXiv
Wu X, Zhong Y, Wu J, et al (2023) AS-LLM: When algorithm selection meets large language model. arXiv
Wu X, Wu SH, Wu J, et al (2024) Evolutionary computation in the era of large language model: survey and roadmap. IEEE Trans Evol Comput 1
Wu X, Shen Y, Ge F, et al (2025) A comprehensive analysis on LLM-based node classification algorithms. arXiv
Xiao L, Chen X, Shan X (2024) Enhancing large language models with evolutionary fine-tuning for news summary generation. J Intell Fuzzy Syst 1–13
Xiao Y, Zhu Y, Qu Y, et al (2025) A market for power system resilience provision. Appl Energy 125335
Yan L, Zhou J, Yang K (2024) Control-aware trajectory predictions for communication-efficient drone swarm coordination in cluttered environments. arXiv
Yan Y, Zhu Y, Xu W (2025) Bias in decision-making for ai's ethical dilemmas: a comparative study of chatgpt and claude. arXiv
Yang XS, Deb S (2009) Cuckoo search via lévy flights. In: 2009 World congress on nature & biologically inspired computing (NaBIC), IEEE, pp 210–214
Yang X, Wang R, Li K, et al (2025a) Meta-black-box optimization for evolutionary algorithms: review and perspective. Swarm Evol Comput 101838
Yang X, Wang R, Li K, et al (2025b) Platmetax: an integrated MATLAB platform for meta-black-box optimization. arXiv
Yang X, Wang R, Li K, et al (2025c) Reinforcement learning based automated design of differential evolution algorithm for black-box optimization. arXiv
Yang XW, Shao JJ, Guo LZ, et al (2025d) Neuro-symbolic artificial intelligence: towards improving the reasoning abilities of large language models. arXiv
Ye H, Wang J, Cao Z, et al (2024) Reevo: large language models as hyper-heuristics with reflective evolution. In: 38th Conference on Neural Information Processing Systems, NeurIPS 2024
Yin F, Zhang Y, Wu B, et al (2024) Generalizable black-box adversarial attack with meta learning. IEEE transactions on pattern analysis and machine intelligence, pp 1804–1818
Yu X, Gen M (2010) Introduction to evolutionary algorithms. Springer, New York
Yu H, Liu J (2024) Deep insights into automated optimization with large language models and evolutionary algorithms. arXiv
Yuksekgonul M, Bianchi F, Boen J et al (2025) Optimizing generative ai by backpropagating language model feedback. Nature 639:609–616
Yun T, Lee K, Yun S, et al (2024) An offline meta black-box optimization framework for adaptive design of urban traffic light management systems. In: KDD '24: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining
Zamfirescu-Pereira JD, Wong RY, Hartmann B, et al (2023) Why johnny can't prompt: how non-ai experts try (and fail) to design llm prompts. In: Proceedings of the 2023 CHI conference on human factors in computing systems, pp 1–21
Zhang J, Arawjo I (2024) Chainbuddy: an AI agent system for generating llm pipelines. arXiv
Zhang Z, Wu Z, Zhang H et al (2022) Meta-learning-based deep reinforcement learning for multiobjective optimization problems. IEEE Trans Neural Netw Learn Syst 34(10):7978–7991
Zhang L, Ergen T, Logeswaran L, et al (2024a) Sprig: improving large language model performance by system prompt optimization. arXiv
Zhang T, Yuan J, Avestimehr S (2024b) Revisiting OPRO: the limitations of small-scale LLMS as optimizers. arXiv
Zhang G, Chen K, Wan G, et al (2025a) Evoflow: evolving diverse agentic workflows on the fly. arXiv
Zhang Z, Li X, Li RH, et al (2025b) Toward general and robust LLM-enhanced text-attributed graph learning. arXiv
Zhao Q, Duan Q, Yan B, et al (2023a) Automated design of metaheuristic algorithms: a survey. arXiv
Zhao WX, Zhou K, Li J, et al (2023b) A survey of large language models. arXiv
Zhao Q, Yan B, Hu T, et al (2025a) AutoOpt: a general framework for automatically designing metaheuristic optimization algorithms with diverse structures. IEEE Trans Emerg Topics Comput Intell 3690–3703
Zhao Z, Hua C, Berto F, et al (2025b) Trajevo: designing trajectory prediction heuristics via LLM-driven evolution. arXiv
Zhong J, Dong J, Liu WL, et al (2025) Multiform genetic programming framework for symbolic regression problems. IEEE Trans Evol Comput 1
Zhou F, Zhang L, Wei W (2022) Meta-generating deep attentive metric for few-shot classification. IEEE Trans Circ Syst Video Technol 6863–6873
Zhou Y, Lei L, Zhao X, et al (2024) Decomposition and meta-drl based multi-objective optimization for asynchronous federated learning in 6g-satellite systems. IEEE J Selected Areas Commun 1115–1129

Zhu Y, Lu H, Wu Y, et al (2025) Constrained offline black-box optimization via risk evaluation and management. In: Proceedings of the AAAI Conference on Artificial Intelligence, pp 23063–23071
Zimmerman RD, Murillo-Sánchez CE, Thomas RJ (2010) Matpower: steady-state operations, planning, and analysis tools for power systems research and education. IEEE Trans Power Syst 26(1):12–19
Zitzler E, Laumanns M, Thiele L (2001) SPEA2: improving the strength pareto evolutionary algorithm. TIK report 103
Zong Y, Wan J (2025) Hierarchical optimization algorithm for the automatic design of analog integrated circuit. In: 2025 Conference of science and technology of integrated circuits (CSTIC)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Yan Zheng[1] · Lida Zhang[1] · Kaiwen Li[1] · Rui Wang[1] · Wenhua Li[1] · Tao Zhang[1] · Qingfu Zhang[2] · Yaochu Jin[3]

✉ Kaiwen Li
kaiwenli_nudt@foxmail.com

✉ Rui Wang
ruiwangnudt@gmail.com

Yan Zheng
zhengyan24@nudt.edu.cn

Lida Zhang
z674737109@163.com

Wenhua Li
liwenhua1030@aliyun.com

Tao Zhang
zhangtao@nudt.edu.cn

Qingfu Zhang
qingfu.zhang@cityu.edu.hk

Yaochu Jin
jinyaochu@westlake.edu.cn

[1] College of Systems Engineering, National University of Defense Technology, Changsha 410073, China

[2] Department of Computer Science, City University of Hong Kong, Hong Kong 999077, China

[3] School of Engineering, Westlake University, Hangzhou 310030, China